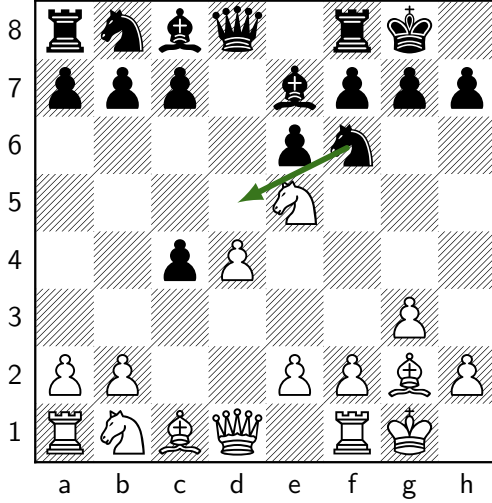


Melanie-XRAY — Catalan

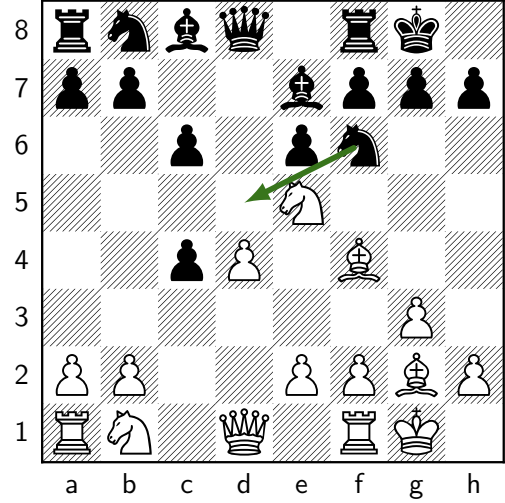
Melanie (Deep_Vision)

Abstract

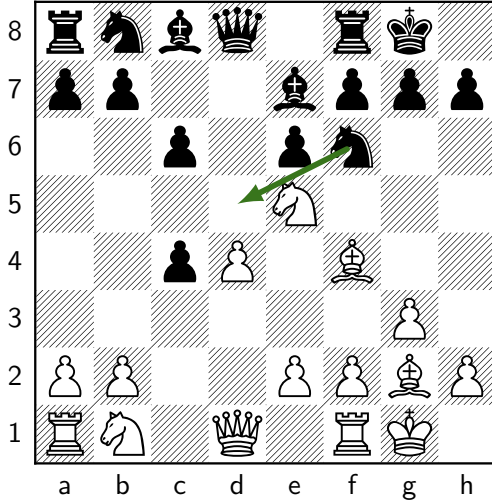
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



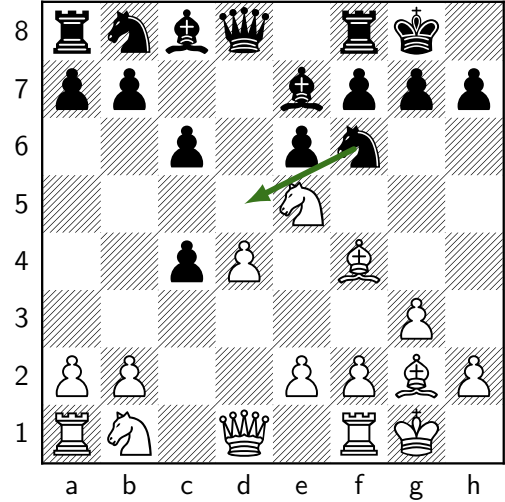
(a)



(b)

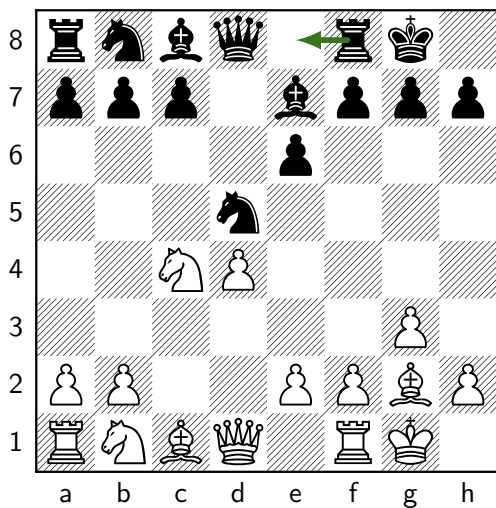


(c)

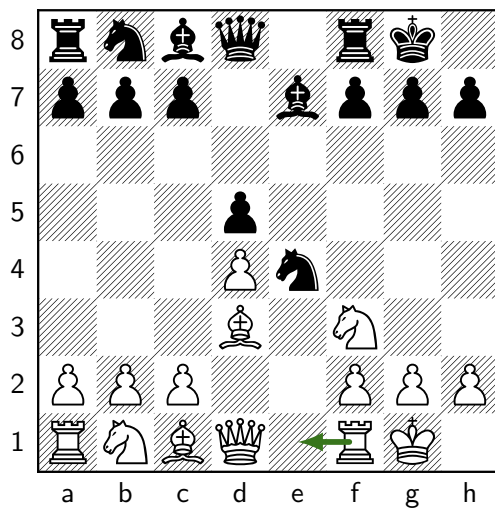


(d)

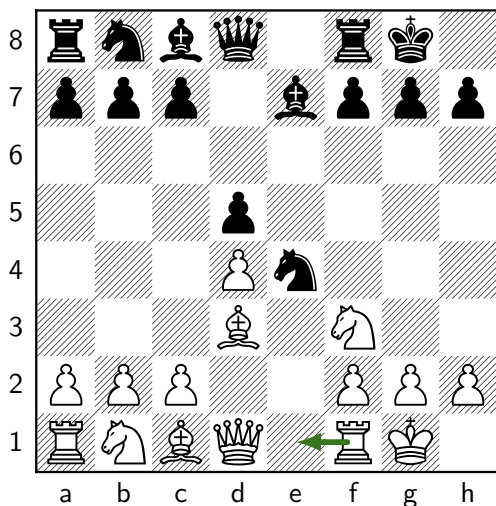
Figure 1: **Move 7, Black: Nd5.** (a) Our position; Melanie plays Nd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#271 (ply 17) [structural distance $d=26$, QAP obj=161]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#272 (ply 17) [structural distance $d=26$, QAP obj=161]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3736 (ply 17) [structural distance $d=26$, QAP obj=161]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd5 (summed QAP objective 964, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



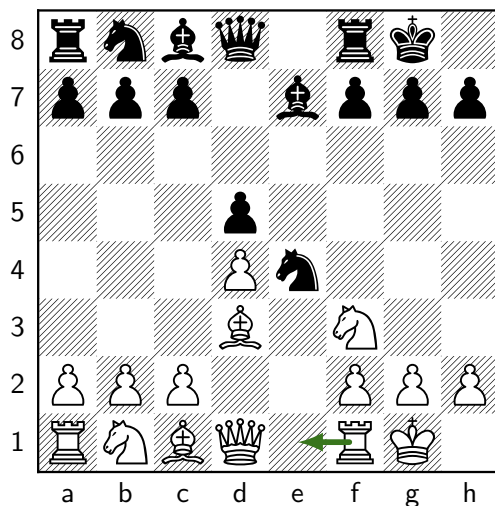
(a)



(b)

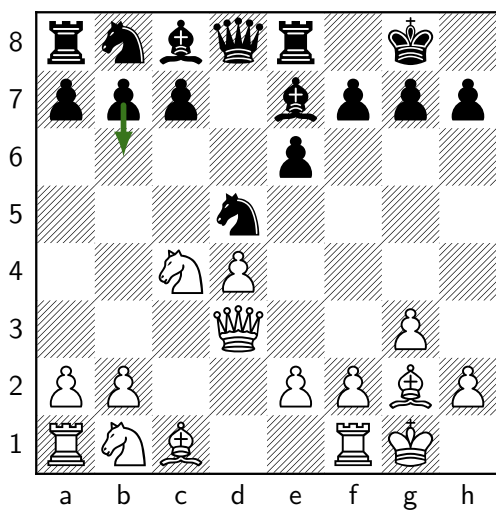


(c)

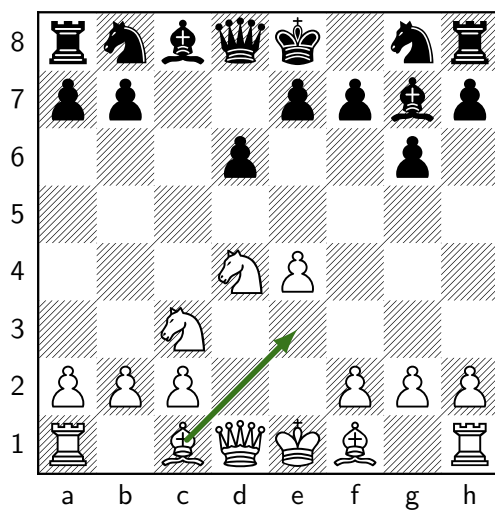


(d)

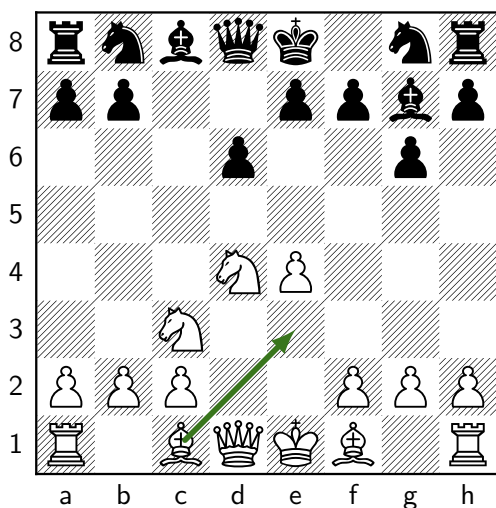
Figure 2: **Move 8, Black: Re8**. (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#629 (ply 14) [structural distance $d=46$, QAP obj=16]: the master played Re1; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pb7 as their Ra1, our ph2 as their Nb1; transferred move f8e8 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#630 (ply 14) [structural distance $d=46$, QAP obj=16]: the master played Re1; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pb7 as their Ra1, our ph2 as their Nb1; transferred move f8e8 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1788 (ply 14) [structural distance $d=46$, QAP obj=16]: the master played Re1; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pb7 as their Ra1, our ph2 as their Nb1; transferred move f8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Re8 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



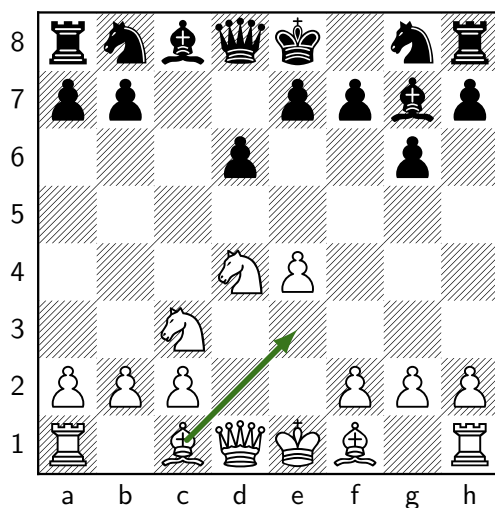
(a)



(b)

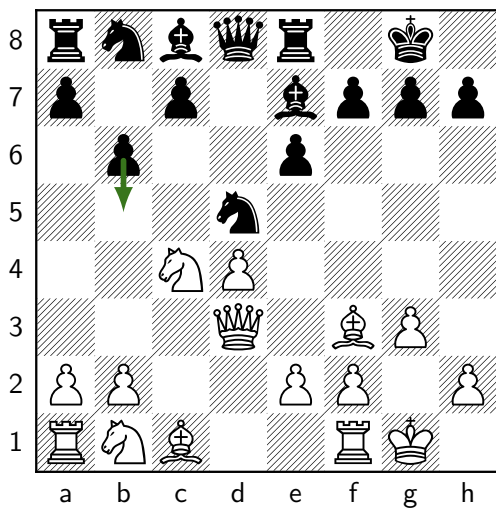


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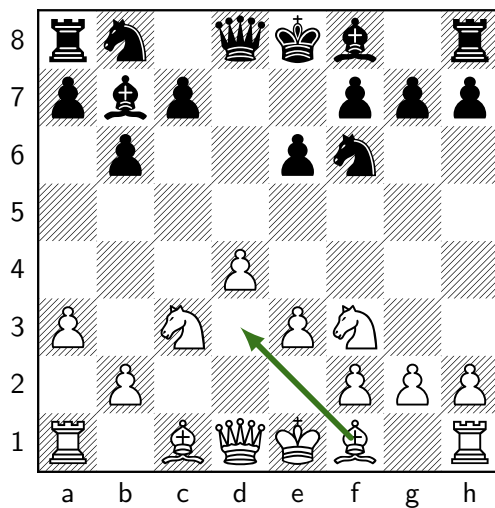


(d)

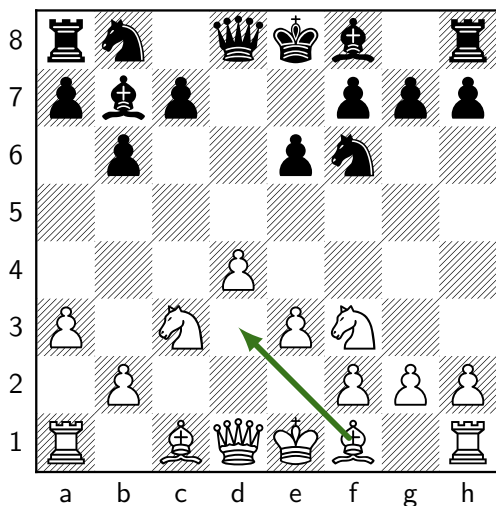
Figure 3: **Move 9, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1092` (ply 10) [structural distance $d=62$, QAP obj=26]: the master played Be3; under μ our *Pb2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Re8* as their *Qd1*, our *Pf7* as their *Bf1*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1093` (ply 10) [structural distance $d=62$, QAP obj=26]: the master played Be3; under μ our *Pb2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Re8* as their *Qd1*, our *Pf7* as their *Bf1*; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1116` (ply 10) [structural distance $d=62$, QAP obj=26]: the master played Be3; under μ our *Pb2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Re8* as their *Qd1*, our *Pf7* as their *Bf1*; transferred move b7b6 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to b6 (summed QAP objective 339, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



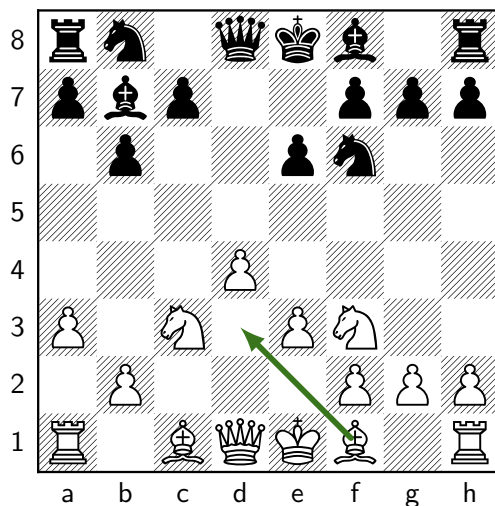
(a)



(b)

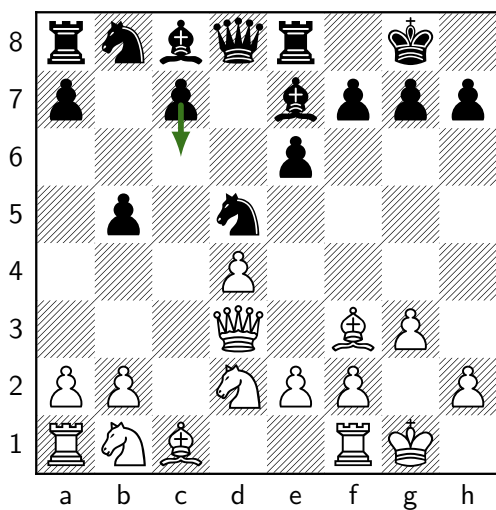


(c)

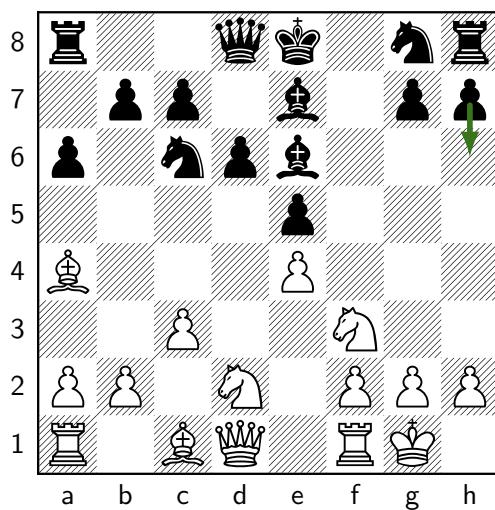


(d)

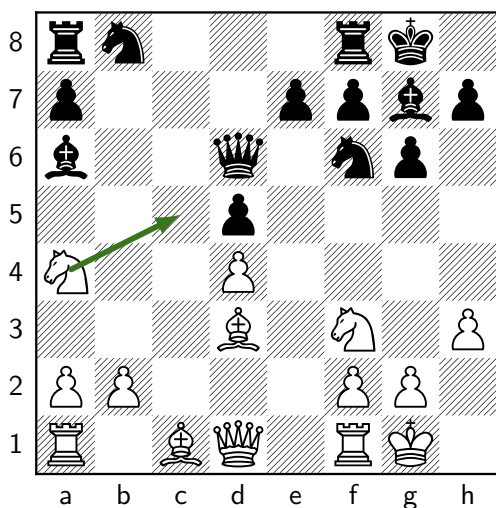
Figure 4: **Move 10, Black: b5.** (a) Our position; Melanie plays b5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1883` (ply 14) [structural distance $d=63$, QAP obj=16]: the master played Bd3; under μ our *Pb3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra1*, our *nc4* as their *Bc1*; transferred move b6b5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1884` (ply 14) [structural distance $d=63$, QAP obj=16]: the master played Bd3; under μ our *Pb3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra1*, our *nc4* as their *Bc1*; transferred move b6b5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2029` (ply 14) [structural distance $d=63$, QAP obj=16]: the master played Bd3; under μ our *Pb3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra1*, our *nc4* as their *Bc1*; transferred move b6b5 (**endorses this move**). Synthesis: 25 retrieved analogs transferred to b5 (summed QAP objective 389, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



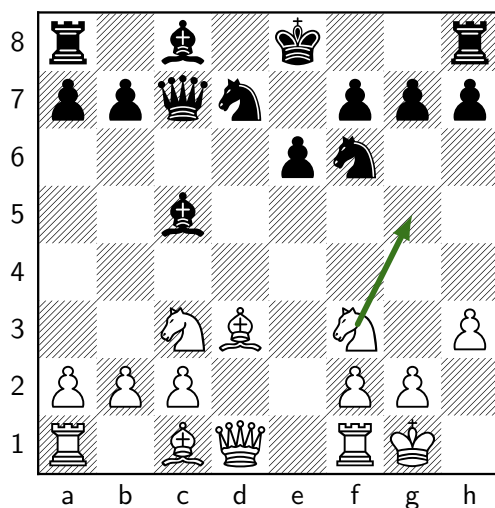
(a)



(b)

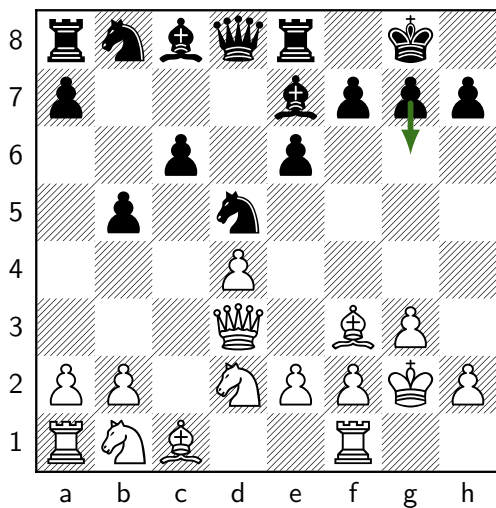


(c)

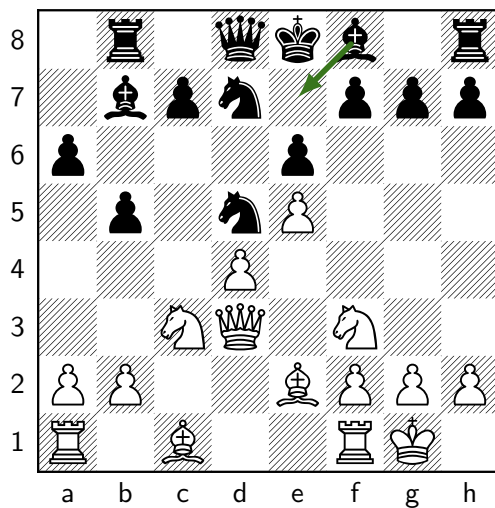


(d)

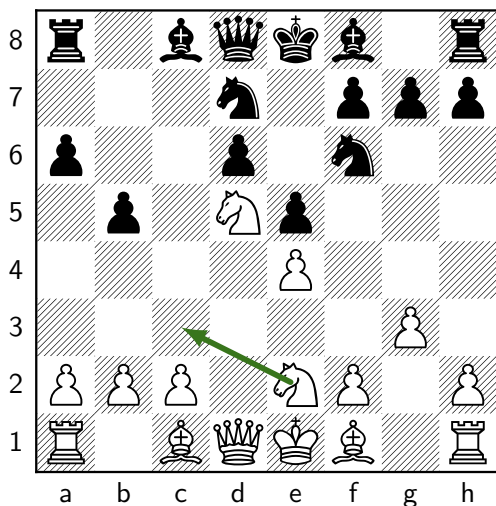
Figure 5: **Move 11, Black: c6**. (a) Our position; Melanie plays c6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1851` (ply 17) [structural distance $d=81$, QAP obj=50]: the master played h6; under μ our *Pc2* plays the role of that game’s *Ph2*; cross-type roles: our *Qd8* as their *Ra8*, our *Re8* as their *Qd8*; transferred move c7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3292` (ply 26) [structural distance $d=72$, QAP obj=39]: the master played Nc5; under μ our *Pc2* plays the role of that game’s *Na4* (a **cross-type** correspondence); cross-type roles: our *Pb5* as their *Ra1*, our *nb1* as their *Bc1*; transferred move c7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2055` (ply 18) [structural distance $d=83$, QAP obj=17]: the master played Ng5; under μ our *Pc2* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *pe2* as their *Bc1*; transferred move c7c6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to c6 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



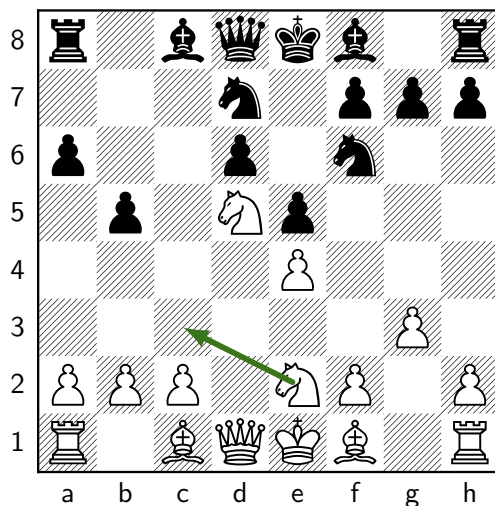
(a)



(b)

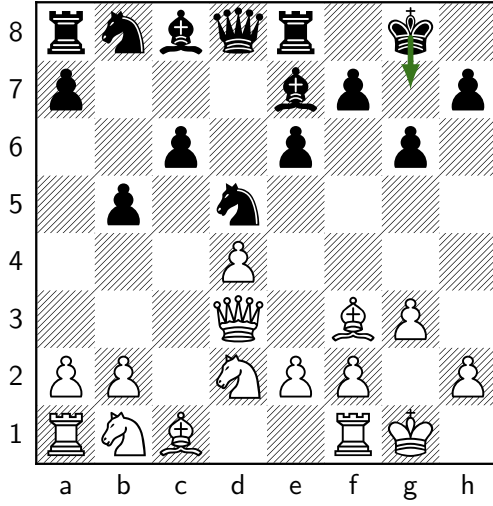


(c)

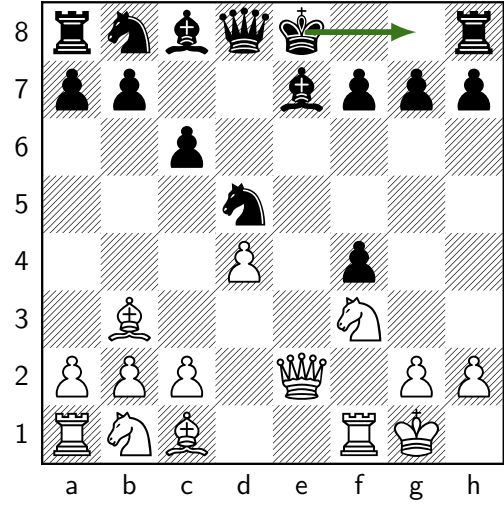


(d)

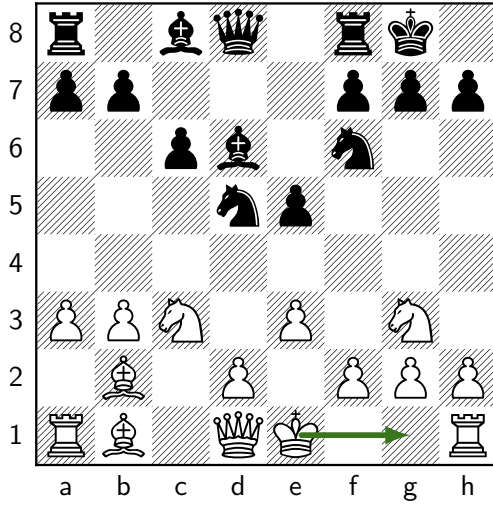
Figure 6: **Move 12, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1342` (ply 21) [structural distance $d=88$, QAP obj=37]: the master played Be7; under μ our Pg2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rb8, our Re8 as their Qd8; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3842` (ply 16) [structural distance $d=81$, QAP obj=30]: the master played Nec3; under μ our Pg2 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Ra1, our nb1 as their Bc1; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3843` (ply 16) [structural distance $d=81$, QAP obj=30]: the master played Nec3; under μ our Pg2 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Ra1, our nb1 as their Bc1; transferred move g7g6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to g6 (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



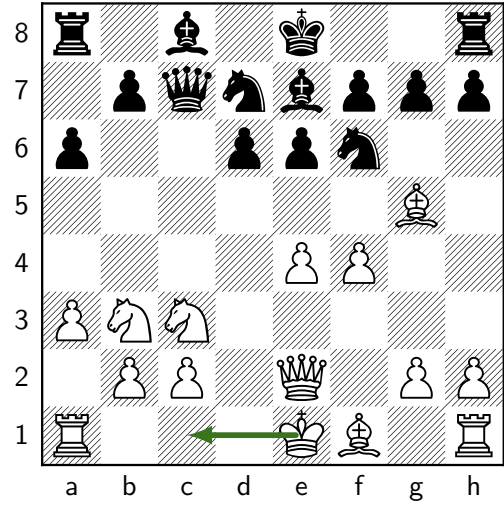
(a)



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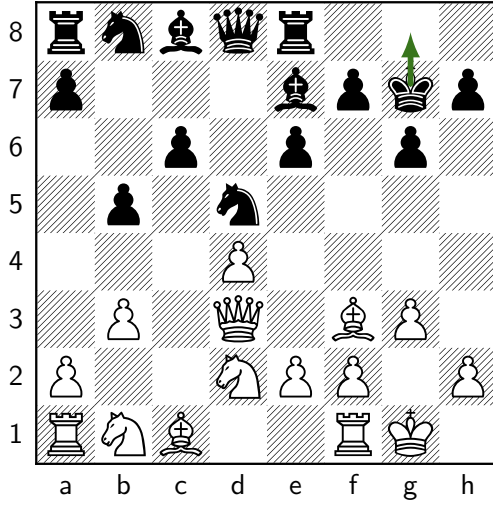


(c)

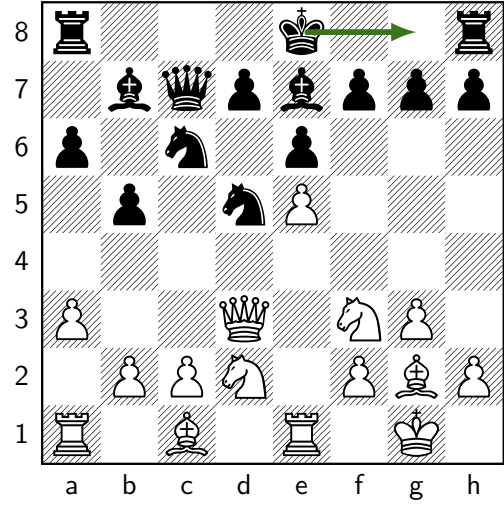


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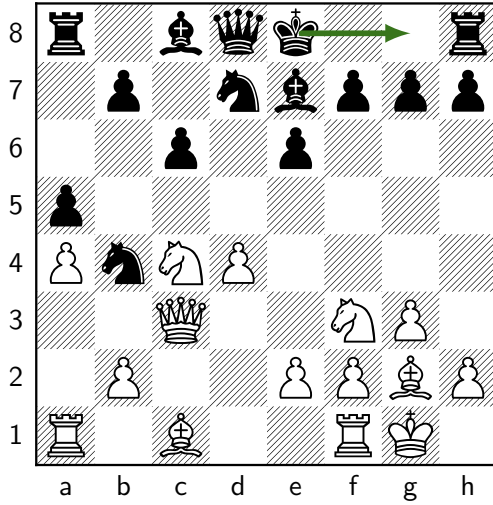
Figure 7: **Move 13, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2753` (ply 17) [structural distance $d=75$, QAP obj=33]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Nb8*, our *pg3* as their *Bc8*; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1482` (ply 22) [structural distance $d=85$, QAP obj=30]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra1*, our *rf1* as their *Qd1*; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1934` (ply 20) [structural distance $d=85$, QAP obj=16]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Ph7* as their *Bf1*, our *Qd8* as their *Rh1*; transferred move g8g7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg7 (summed QAP objective 170, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



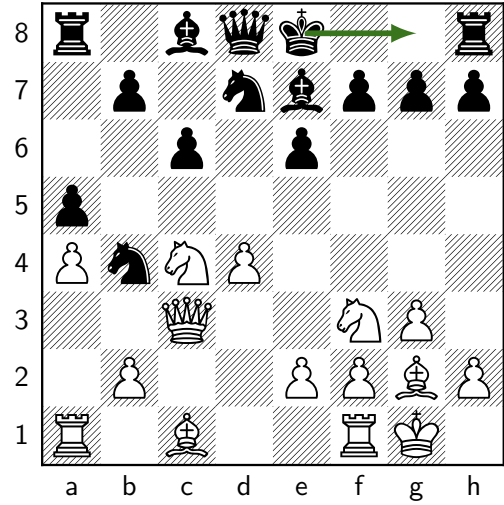
(a)



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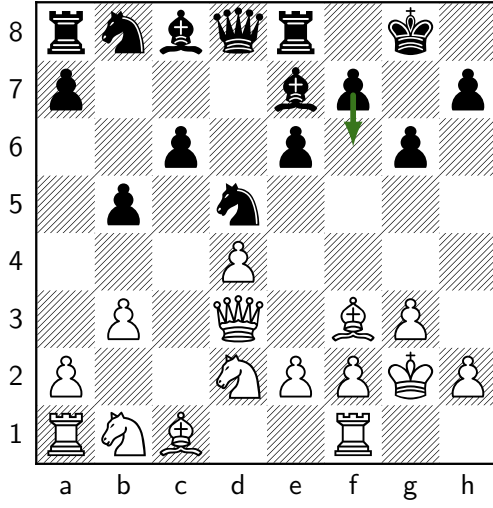


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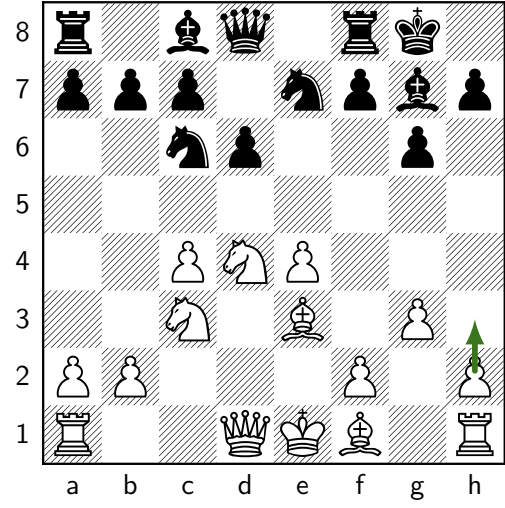


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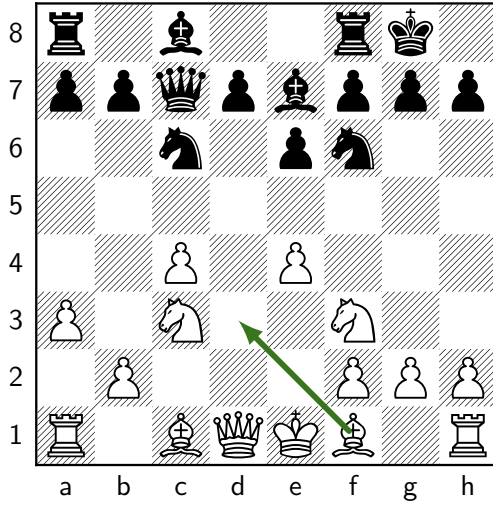
Figure 8: **Move 14, Black: Kg8**. (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3349` (ply 23) [structural distance $d=82$, QAP obj=38]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *pe2* as their *Bb7*; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2993` (ply 21) [structural distance $d=86$, QAP obj=17]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move g7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2994` (ply 21) [structural distance $d=86$, QAP obj=17]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move g7g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



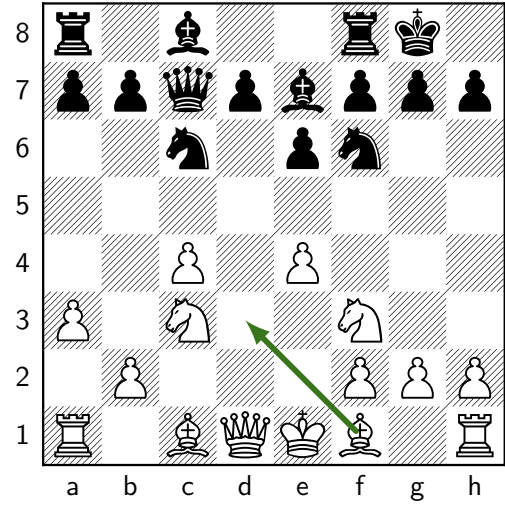
(a)



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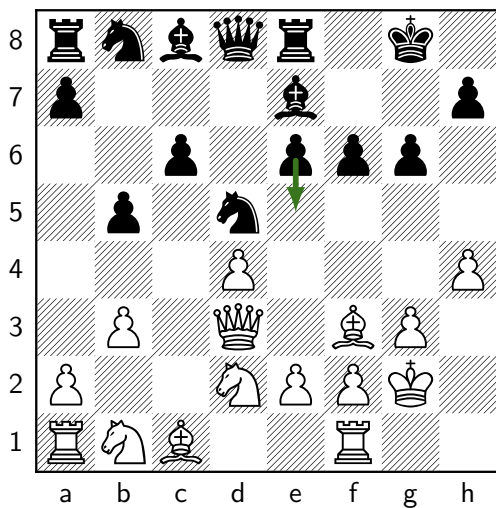


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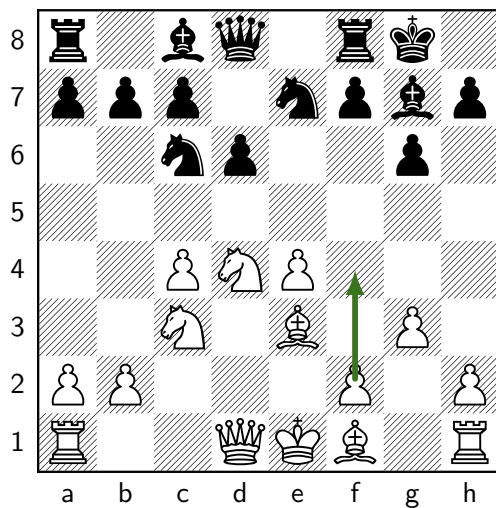


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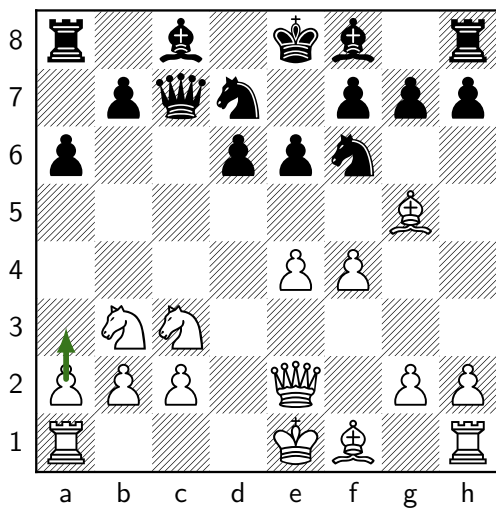
Figure 9: **Move 15, Black: f6.** (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1598` (ply 16) [structural distance $d=95$, QAP obj=37]: the master played h3; under μ our *Pf2* plays the role of that game’s *Ph2*; cross-type roles: our *Qd8* as their *Ra1*, our *Re8* as their *Qd1*; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#93` (ply 16) [structural distance $d=99$, QAP obj=14]: the master played Bd3; under μ our *Pf2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra1*, our *Pb5* as their *Bc1*; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#94` (ply 16) [structural distance $d=99$, QAP obj=14]: the master played Bd3; under μ our *Pf2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra1*, our *Pb5* as their *Bc1*; transferred move f7f6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to f6 (summed QAP objective 193, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



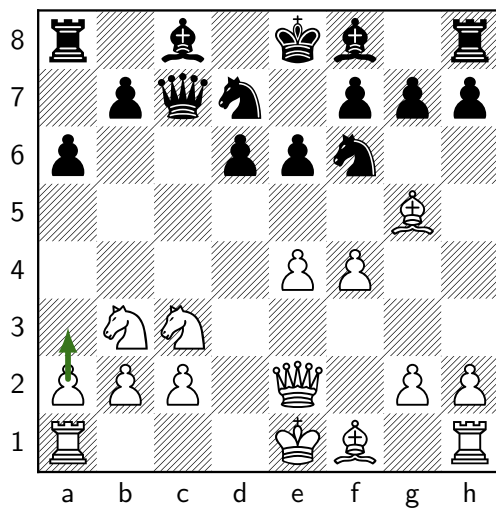
(a)



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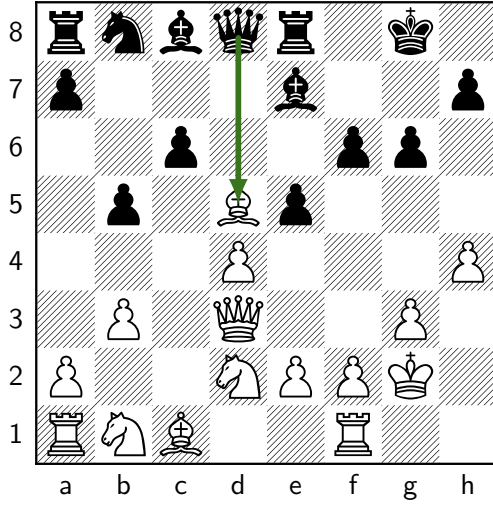


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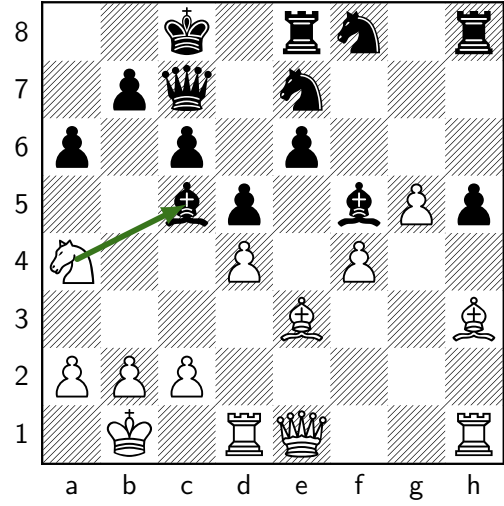


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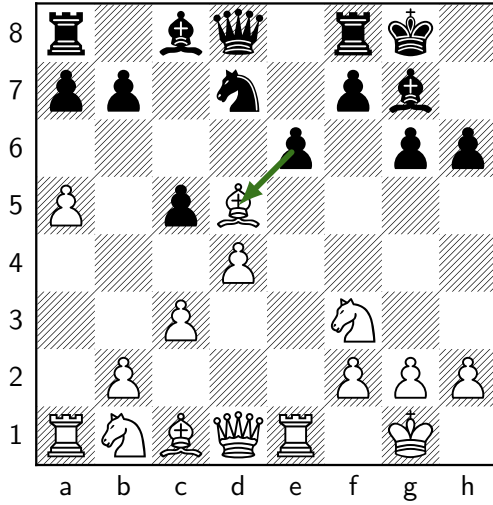
Figure 10: **Move 16, Black: e5 (e6e5).** (a) Our position; Melanie plays e5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1597` (ply 16) [structural distance $d=95$, QAP obj=36]: the master played f4; under μ our $Pe3$ plays the role of that game’s $Pf2$; cross-type roles: our $Qd8$ as their $Ra1$, our $Re8$ as their $Qd1$; transferred move e6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1934` (ply 18) [structural distance $d=85$, QAP obj=15]: the master played a3; under μ our $Pe3$ plays the role of that game’s $Pa2$; cross-type roles: our $Pc6$ as their $Bf1$, our $Qd8$ as their $Rh1$; transferred move e6e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1992` (ply 18) [structural distance $d=85$, QAP obj=15]: the master played a3; under μ our $Pe3$ plays the role of that game’s $Pa2$; cross-type roles: our $Pc6$ as their $Bf1$, our $Qd8$ as their $Rh1$; transferred move e6e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to e5 (summed QAP objective 98, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



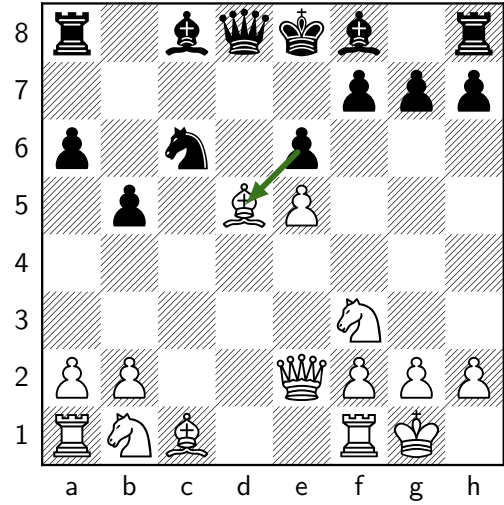
(a)



(b)

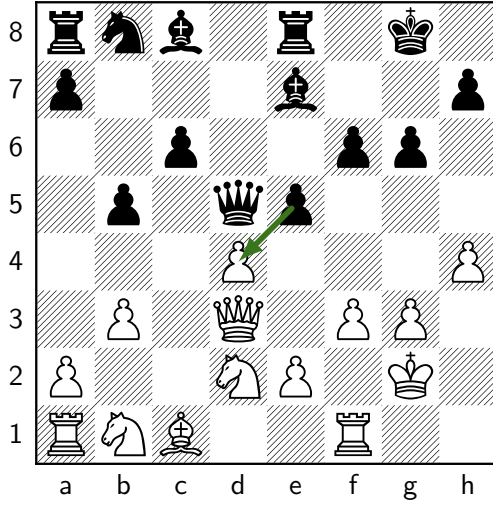


(c)

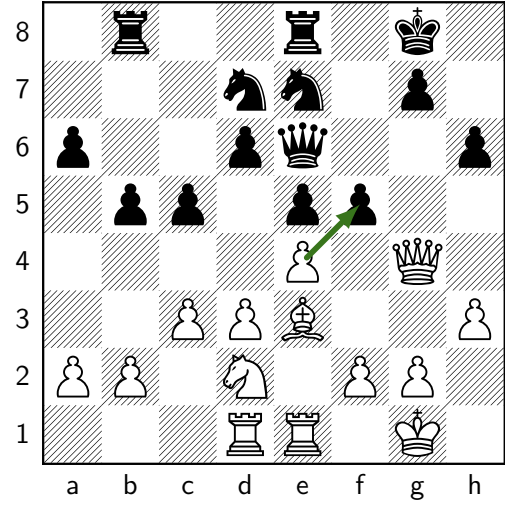


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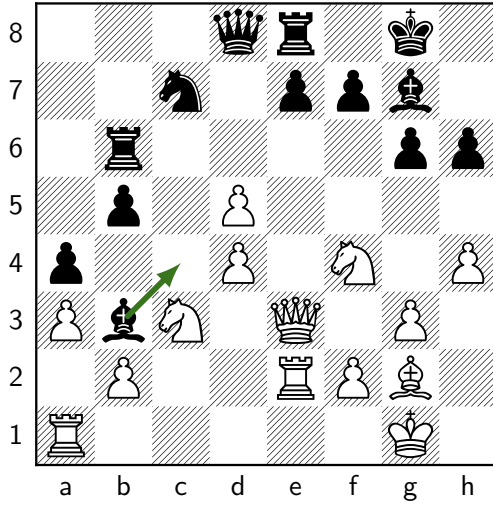
Figure 11: **Move 17, Black: Qxd5+.** (a) Our position; Melanie plays Qxd5+ (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3315` (ply 42) [structural distance $d=110$, QAP obj=35]: the master played Nxc5; under μ our Qd1 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Qe1, our Be7 as their Rh1; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#348` (ply 25) [structural distance $d=110$, QAP obj=24]: the master played exd5; under μ our Qd1 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Ra8, our Nb8 as their Bc8; transferred move d8d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2203` (ply 21) [structural distance $d=115$, QAP obj=24]: the master played exd5; under μ our Qd1 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Re8 as their Bc8, our ph4 as their Bf8; transferred move d8d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qxd5+ (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



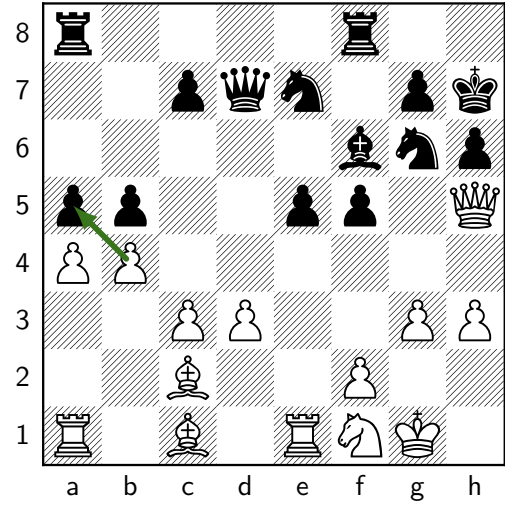
(a)



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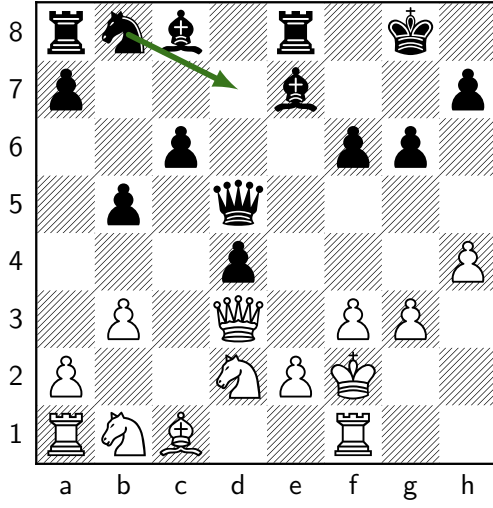


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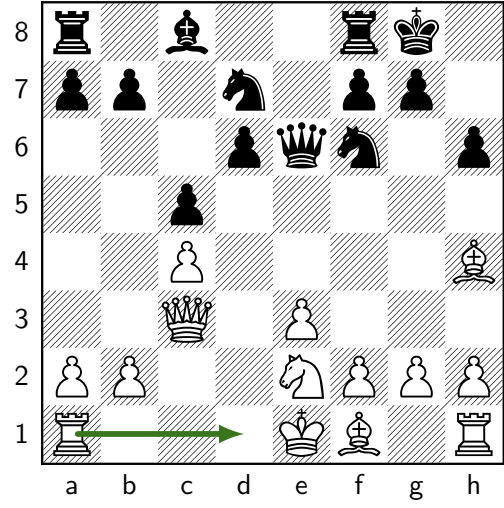


(d)

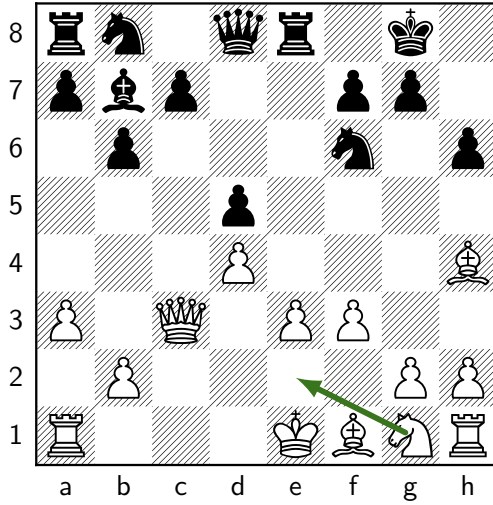
Figure 12: **Move 18, Black: exd4.** (a) Our position; Melanie plays exd4 (e5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3467 (ply 40) [structural distance $d=123$, QAP obj=95]: the master played exf5; under μ our Pe_4 plays the role of that game’s Pe_4 ; cross-type roles: our Be_7 as their Rd_1 , our Nb_8 as their Pg_2 ; transferred move e5d4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#288 (ply 45) [structural distance $d=124$, QAP obj=73]: the master played Bc4; under μ our Pe_4 plays the role of that game’s Bb_6 (a **cross-type** correspondence); cross-type roles: our Pe_5 as their Nc_7 , our Qd_5 as their Bg_7 ; transferred move e5d4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#746 (ply 48) [structural distance $d=124$, QAP obj=55]: the master played bxa5; under μ our Pe_4 plays the role of that game’s Pb_4 ; cross-type roles: our Pc_6 as their Ra_1 , our Bc_8 as their Re_1 ; transferred move e5d4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to exd4 (summed QAP objective 415, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



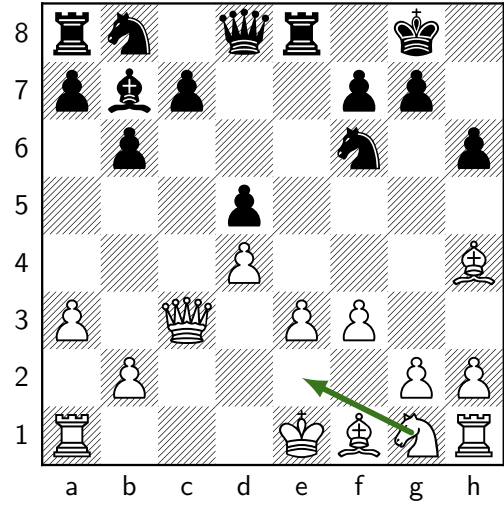
(a)



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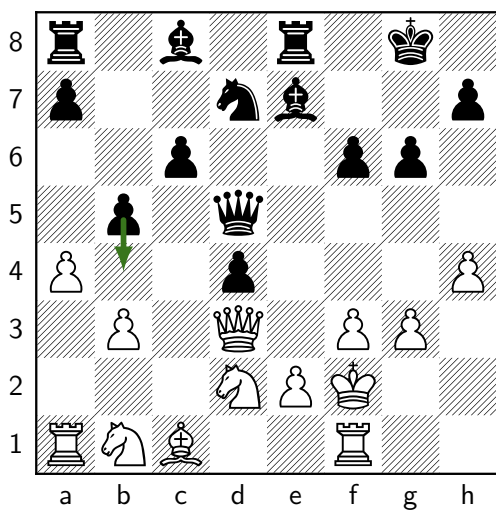


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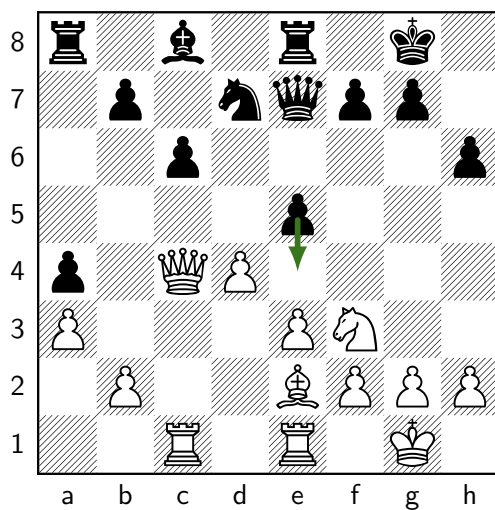


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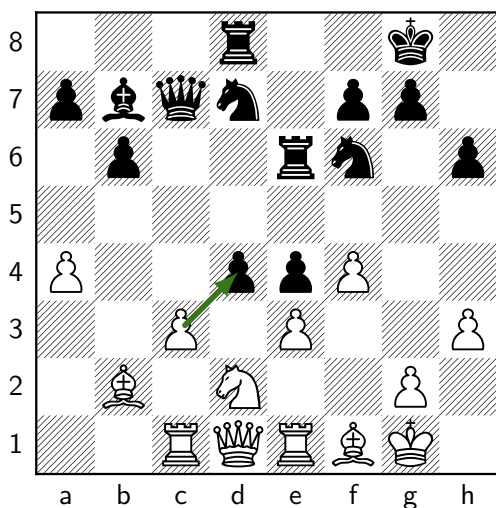
Figure 13: **Move 19, Black: Nd7.** (a) Our position; Melanie plays Nd7 (b8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1200` (ply 22) [structural distance $d=76$, QAP obj=52]: the master played Rd1; under μ our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our ph4 as their Bf1, our Bc8 as their Pa2; transferred move b8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#396` (ply 22) [structural distance $d=99$, QAP obj=34]: the master played Ne2; under μ our Nb1 plays the role of that game’s Ng1; cross-type roles: our pb3 as their Ra1, our Pg6 as their Bf1; transferred move b8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#397` (ply 22) [structural distance $d=99$, QAP obj=34]: the master played Ne2; under μ our Nb1 plays the role of that game’s Ng1; cross-type roles: our pb3 as their Ra1, our Pg6 as their Bf1; transferred move b8d7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Nd7 (summed QAP objective 288, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



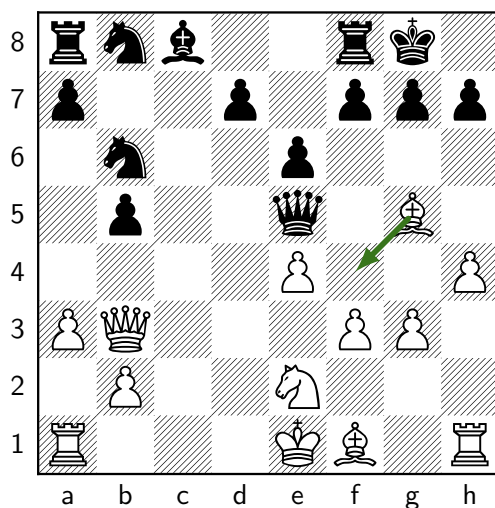
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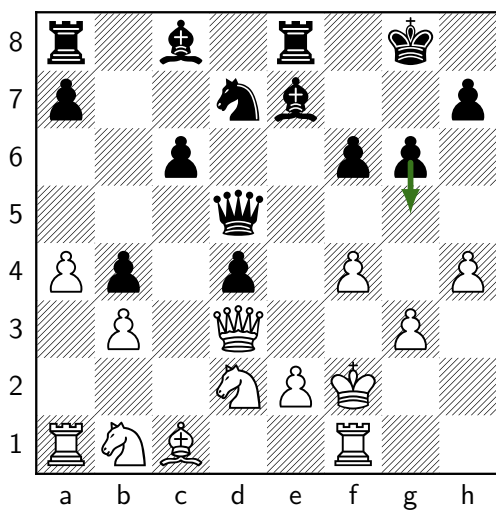


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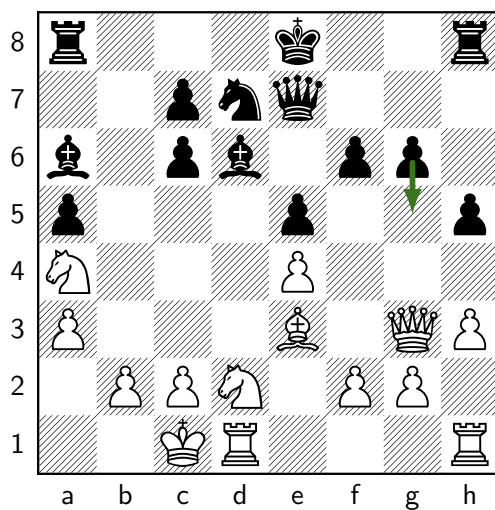


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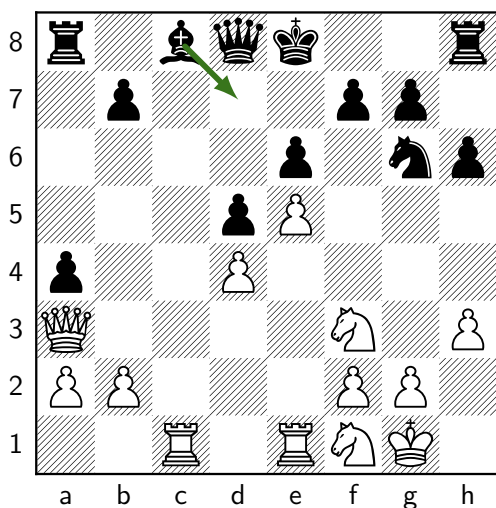
Figure 14: **Move 20, Black: b4**. (a) Our position; Melanie plays b4 (b5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3307` (ply 33) [structural distance $d=112$, QAP obj=123]: the master played e4; under μ our Pb_4 plays the role of that game’s Pe_4 ; cross-type roles: our Be_7 as their Ra_8 , our pg_3 as their Bc_8 ; transferred move b5b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2209` (ply 38) [structural distance $d=98$, QAP obj=111]: the master played cxd4; under μ our Pb_4 plays the role of that game’s Pc_3 ; cross-type roles: our Qd_5 as their Rc_1 , our Ra_8 as their Qd_1 ; transferred move b5b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1938` (ply 28) [structural distance $d=96$, QAP obj=97]: the master played Bf4; under μ our Pb_4 plays the role of that game’s Bg_5 (a **cross-type** correspondence); cross-type roles: our Qd_5 as their Ra_1 , our Ph_7 as their Ne_2 ; transferred move b5b4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to b4 (summed QAP objective 553, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



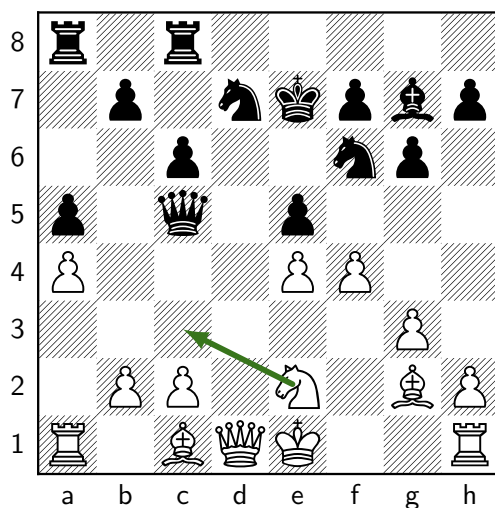
(a)



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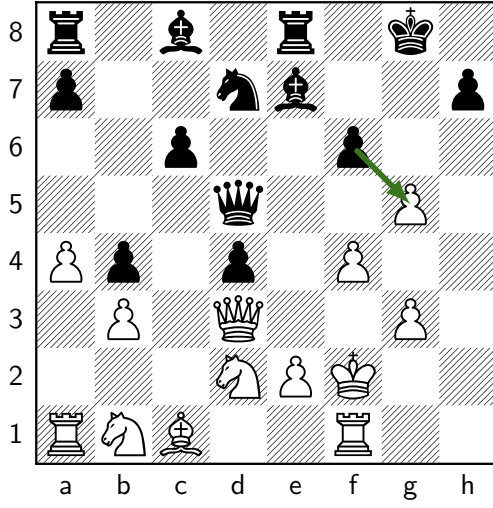


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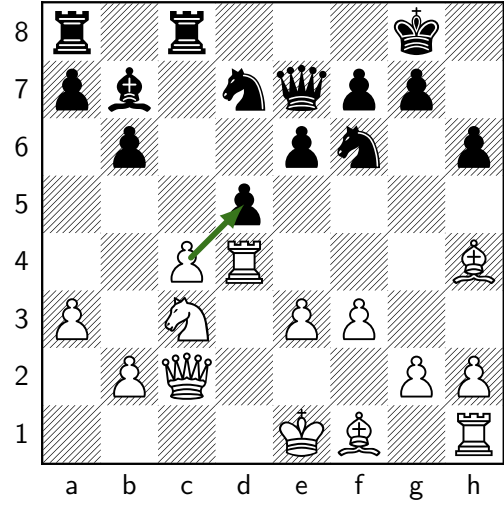


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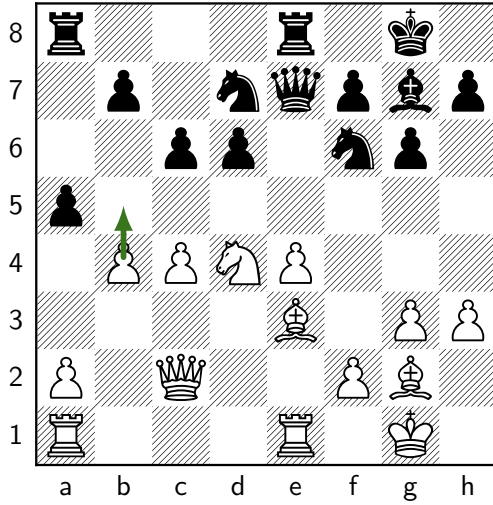
Figure 15: **Move 21, Black: g5**. (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3494` (ply 35) [structural distance $d=112$, QAP obj=65]: the master played g5; under μ our *Pg3* plays the role of that game’s *Pg3*; cross-type roles: our *Be7* as their *Rh8*, our *Ra8* as their *Pc7*; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#674` (ply 37) [structural distance $d=91$, QAP obj=59]: the master played Bd7; under μ our *Pg3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Ra8*, our *Re8* as their *Qd8*; transferred move g6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#410` (ply 28) [structural distance $d=80$, QAP obj=41]: the master played Nc3; under μ our *Pg3* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Pb4* as their *Ra1*, our *pe2* as their *Bc1*; transferred move g6g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g5 (summed QAP objective 197, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



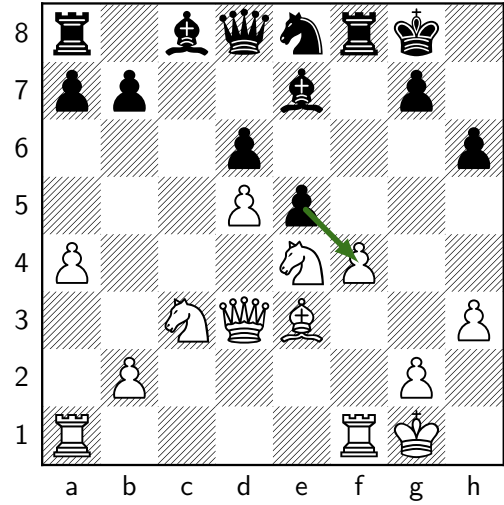
(a)



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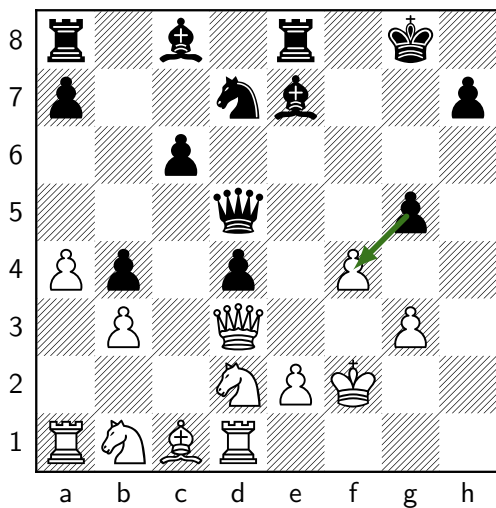


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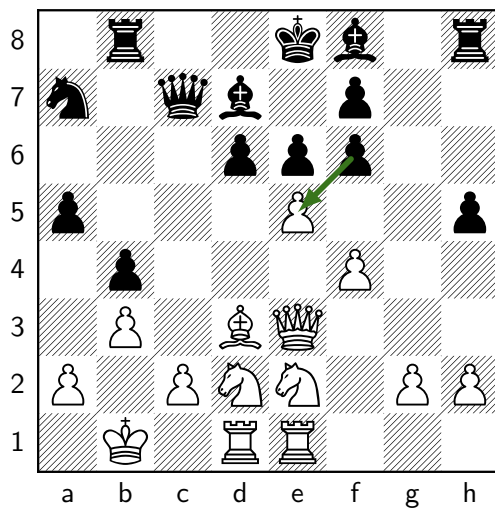


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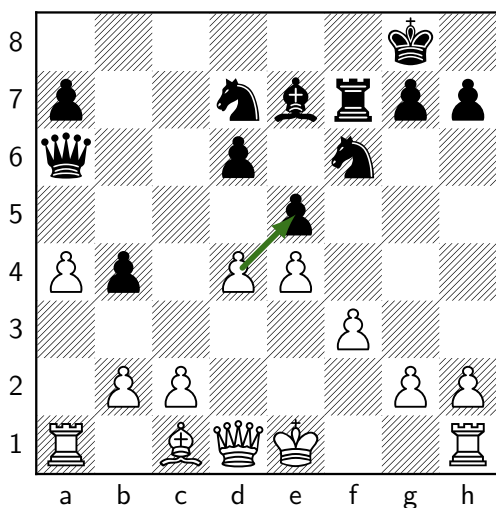
Figure 16: **Move 22, Black: f6g5.** (a) Our position; Melanie plays f6g5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2723 (ply 30) [structural distance $d=114$, QAP obj=81]: the master played cxd5; under μ our Pf3 plays the role of that game’s Pc4; cross-type roles: our ra1 as their Bf1, our Nd7 as their Qc2; transferred move f6g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1067 (ply 38) [structural distance $d=111$, QAP obj=70]: the master played b5; under μ our Pf3 plays the role of that game’s Pb4; cross-type roles: our Pa7 as their Ra1, our Nd7 as their Re1; transferred move f6g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#732 (ply 31) [structural distance $d=112$, QAP obj=63]: the master played exf4; under μ our Pf3 plays the role of that game’s Pe4; cross-type roles: our Pb4 as their Bc8, our Be7 as their Qd8; transferred move f6g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to f6g5 (summed QAP objective 416, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



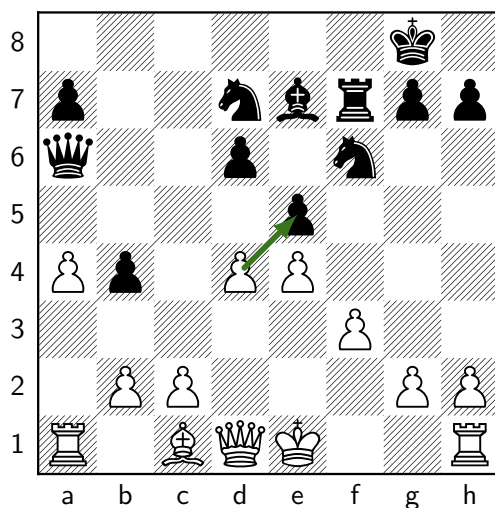
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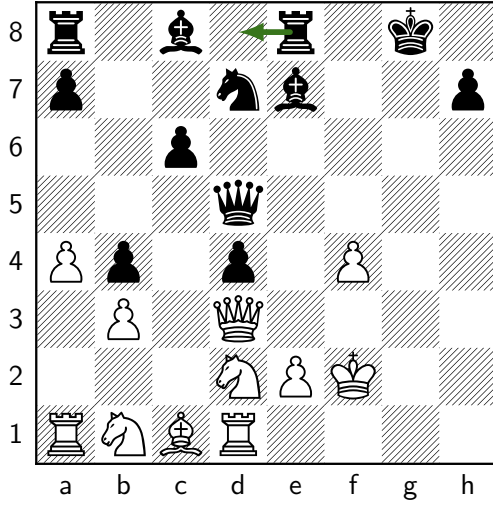


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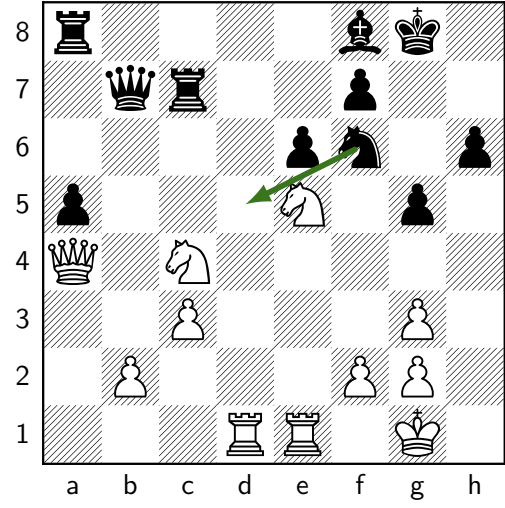


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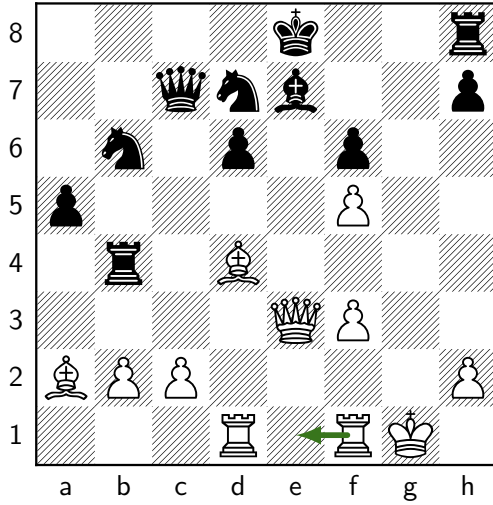
Figure 17: **Move 23, Black: gxf4**. (a) Our position; Melanie plays **gxf4** (**g5f4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#137` (ply 33) [structural distance $d=127$, QAP obj=90]: the master played **fxe5**; under μ our $Pg4$ plays the role of that game’s $Pf3$; cross-type roles: our $Pa7$ as their $Rh8$, our $Pd4$ as their $Na7$; transferred move **g5f4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2508` (ply 30) [structural distance $d=118$, QAP obj=75]: the master played **dxe5**; under μ our $Pg4$ plays the role of that game’s $Pd4$; cross-type roles: our $nb1$ as their $Qd1$, our $nd2$ as their $Ph2$; transferred move **g5f4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#823` (ply 30) [structural distance $d=118$, QAP obj=75]: the master played **dxe5**; under μ our $Pg4$ plays the role of that game’s $Pd4$; cross-type roles: our $nb1$ as their $Qd1$, our $nd2$ as their $Ph2$; transferred move **g5f4** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to **gxf4** (summed QAP objective 583, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



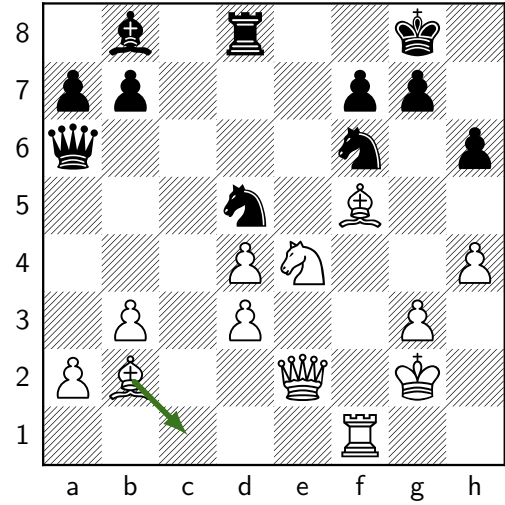
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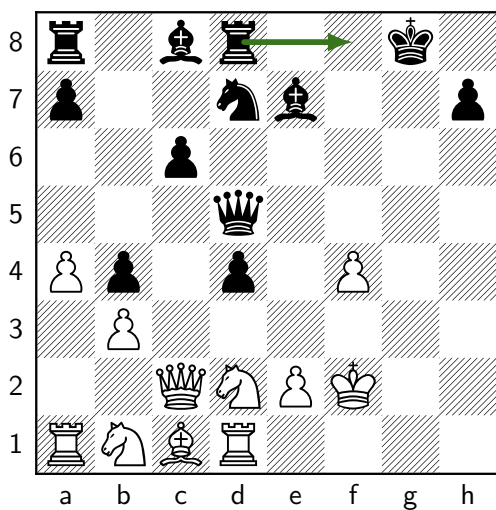


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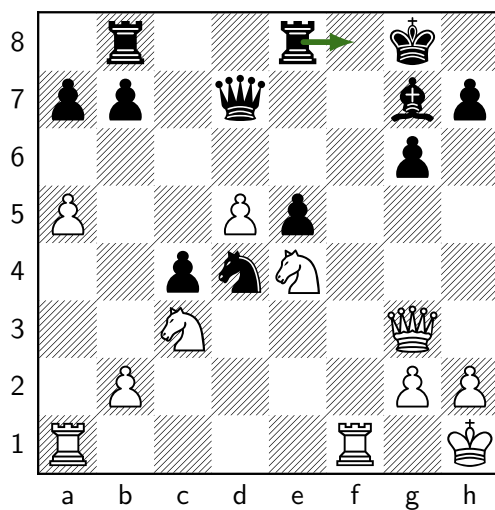


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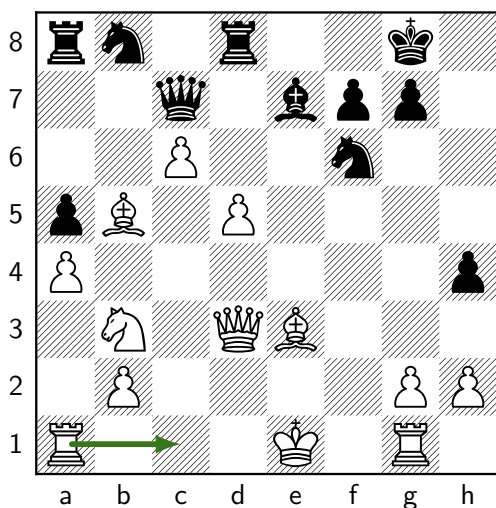
Figure 18: **Move 24, Black: Rd8.** (a) Our position; Melanie plays Rd8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#723` (ply 57) [structural distance $d=136$, QAP obj=64]: the master played Nd5; under μ our *Re1* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Ra8*, our *Ph7* as their *Bf8*; transferred move e8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#957` (ply 52) [structural distance $d=129$, QAP obj=52]: the master played Rfe1; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *pe2* as their *Rd1*, our *nb1* as their *Ba2*; transferred move e8d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#189` (ply 56) [structural distance $d=134$, QAP obj=37]: the master played Bc1; under μ our *Re1* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Pb3*, our *Pd4* as their *Ne4*; transferred move e8d8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rd8 (summed QAP objective 266, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



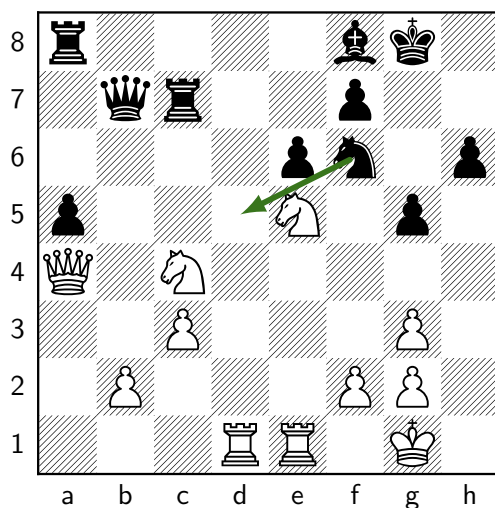
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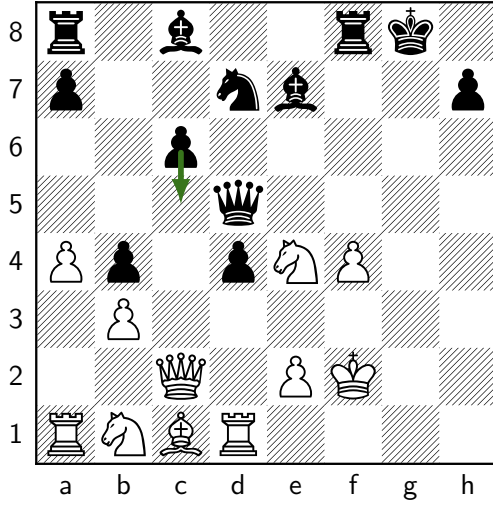


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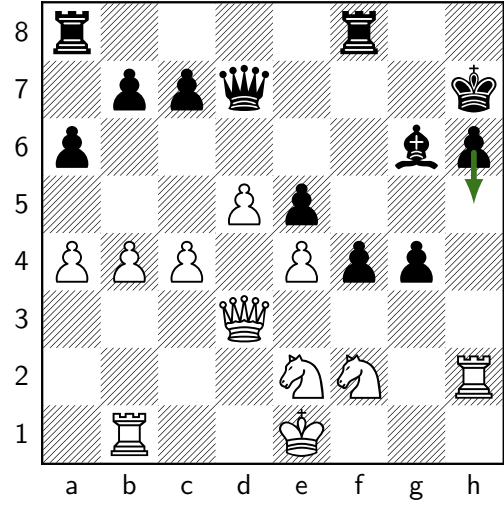


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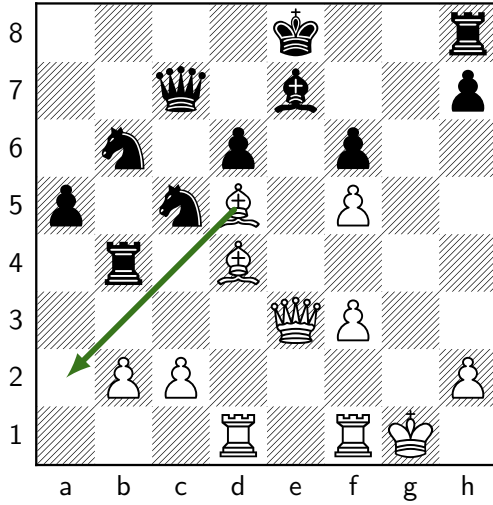
Figure 19: **Move 25, Black: Rf8.** (a) Our position; Melanie plays Rf8 (d8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1352` (ply 55) [structural distance $d=130$, QAP obj=65]: the master played Rf8; under μ our *Rd1* plays the role of that game’s *Re1*; cross-type roles: our *Bc8* as their *Rb8*, our *Be7* as their *Pa7*; transferred move d8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2193` (ply 44) [structural distance $d=111$, QAP obj=39]: the master played Rc1; under μ our *Rd1* plays the role of that game’s *Ra1*; cross-type roles: our *ra1* as their *Ph2*, our *Be7* as their *Qd3*; transferred move d8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#723` (ply 57) [structural distance $d=137$, QAP obj=29]: the master played Nd5; under μ our *Rd1* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Ph7* as their *Bf8*; transferred move d8f8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rf8 (summed QAP objective 133, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



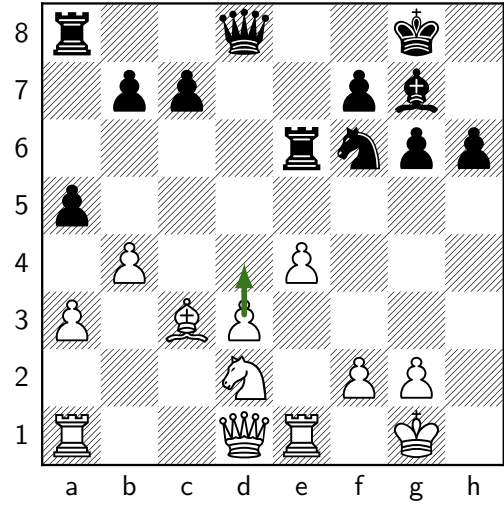
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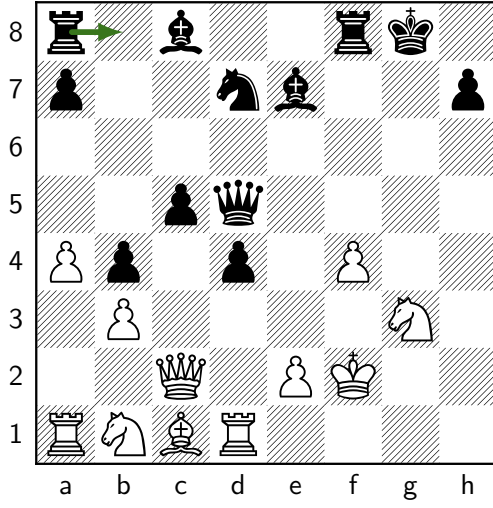


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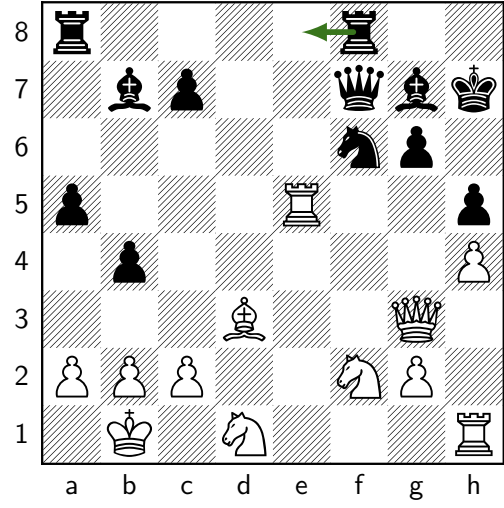


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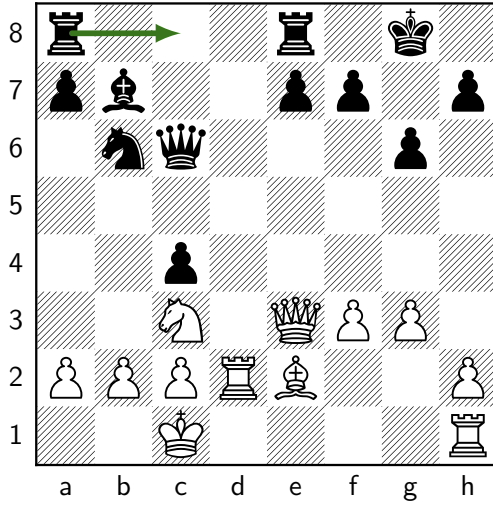
Figure 20: **Move 26, Black: c5.** (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1016` (ply 61) [structural distance $d=98$, QAP obj=46]: the master played h5; under μ our $Pc3$ plays the role of that game’s $Ph3$; cross-type roles: our $Bc8$ as their $Ra8$, our $Pa7$ as their $Rf8$; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#957` (ply 50) [structural distance $d=94$, QAP obj=35]: the master played Ba2; under μ our $Pc3$ plays the role of that game’s $Bd5$ (a **cross-type** correspondence); cross-type roles: our $bc1$ as their $Rd1$, our $Be7$ as their $Pb2$; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3408` (ply 38) [structural distance $d=95$, QAP obj=32]: the master played d4; under μ our $Pc3$ plays the role of that game’s $Pd3$; cross-type roles: our $Bc8$ as their $Ra1$, our $pa4$ as their $Qd1$; transferred move c6c5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to c5 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



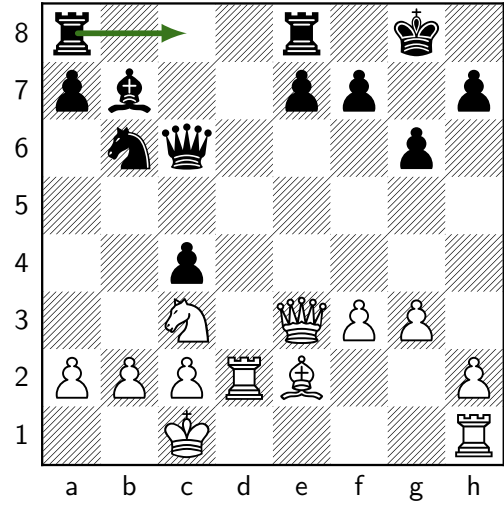
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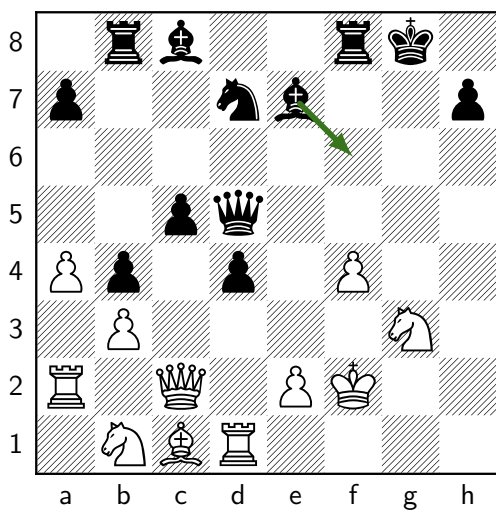


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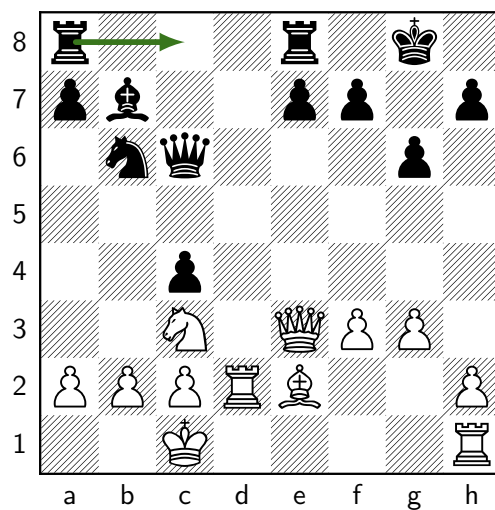


(d)

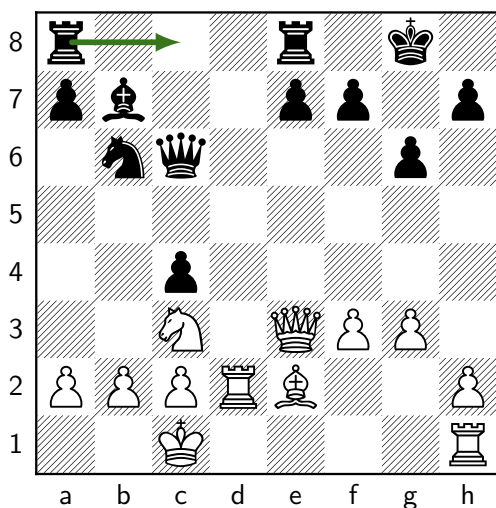
Figure 21: **Move 27, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1169` (ply 51) [structural distance $d=78$, QAP obj=119]: the master played Rfe8; under μ our Ra1 plays the role of that game’s Rf1; cross-type roles: our Pa7 as their Ra8, our Rf8 as their Bb7; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1199` (ply 41) [structural distance $d=67$, QAP obj=67]: the master played Rac8; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our Qd5 as their Re8, our pe2 as their Bb7; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2244` (ply 41) [structural distance $d=67$, QAP obj=67]: the master played Rac8; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our Qd5 as their Re8, our pe2 as their Bb7; transferred move a8b8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rb8 (summed QAP objective 388, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



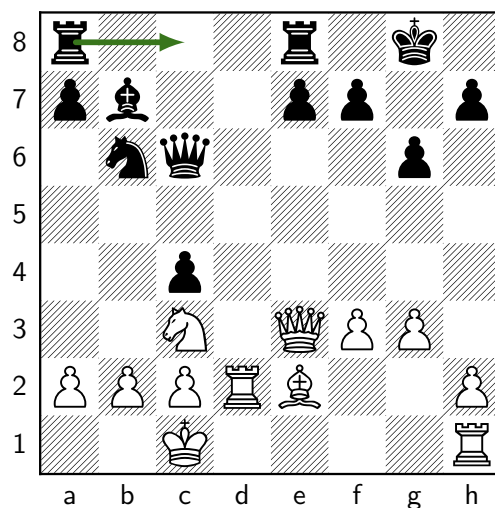
(a)



(b)



(c)



(d)

Figure 22: **Move 28, Black: Bf6**. (a) Our position; Melanie plays Bf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 41) [structural distance $d=80$, QAP obj=57]: the master played Rac8; under μ our Be2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qd5 as their Re8, our ra2 as their Bb7; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2244` (ply 41) [structural distance $d=80$, QAP obj=57]: the master played Rac8; under μ our Be2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qd5 as their Re8, our ra2 as their Bb7; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1599` (ply 41) [structural distance $d=80$, QAP obj=57]: the master played Rac8; under μ our Be2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qd5 as their Re8, our ra2 as their Bb7; transferred move e7f6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf6 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

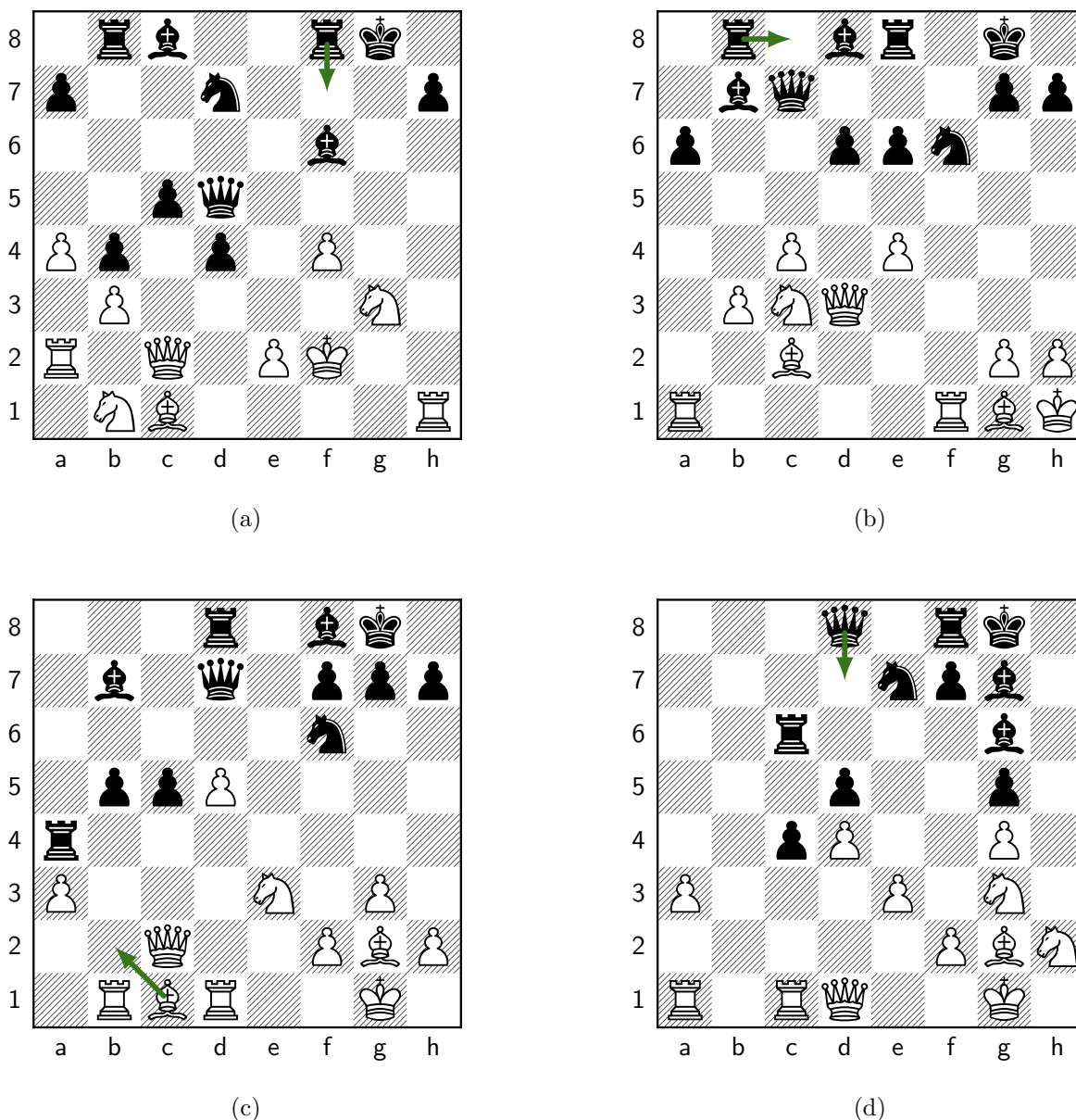
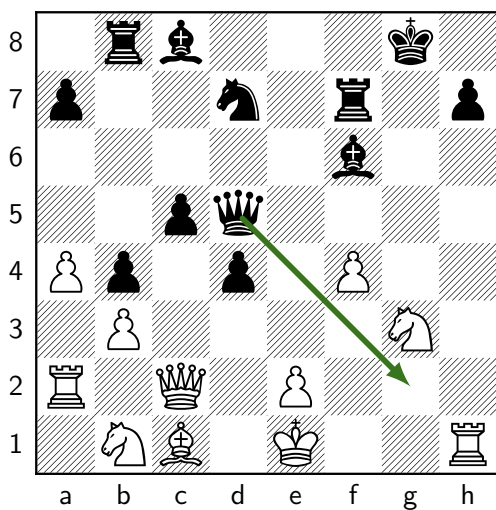
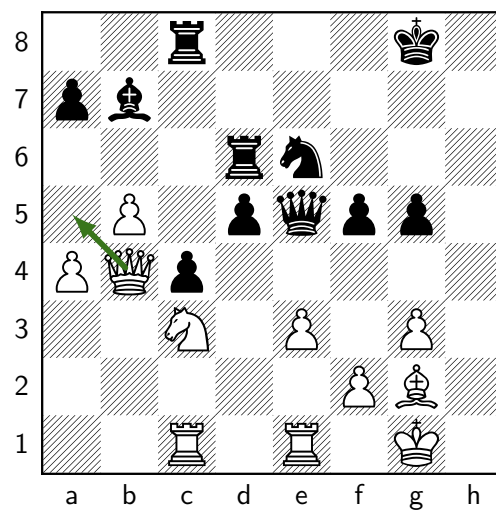


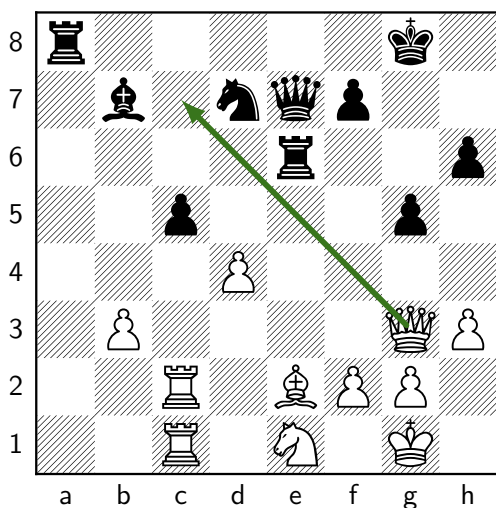
Figure 23: **Move 29, Black: Rf7.** (a) Our position; Melanie plays Rf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2426` (ply 57) [structural distance $d=76$, QAP obj=71]: the master played Rc8; under μ our *Pa2* plays the role of that game’s *Rb1* (a **cross-type** correspondence); cross-type roles: our *pa4* as their *Bd8*, our *Pd4* as their *Bb7*; transferred move a7a6. (c) Retrieved from `ccr1_top_p0003#198` (ply 46) [structural distance $d=90$, QAP obj=66]: the master played Bb2; under μ our *Rf1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *qc2* as their *Rb1*, our *pe2* as their *Rd1*; transferred move f8d8. (d) Retrieved from `ccr1_top_p0002#1535` (ply 53) [structural distance $d=80$, QAP obj=60]: the master played Qd7; under μ our *Rb1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pa4* as their *Ne7*, our *Bf6* as their *Pf7*; transferred move b8a8. Synthesis: no retrieved analog transferred to Rf7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



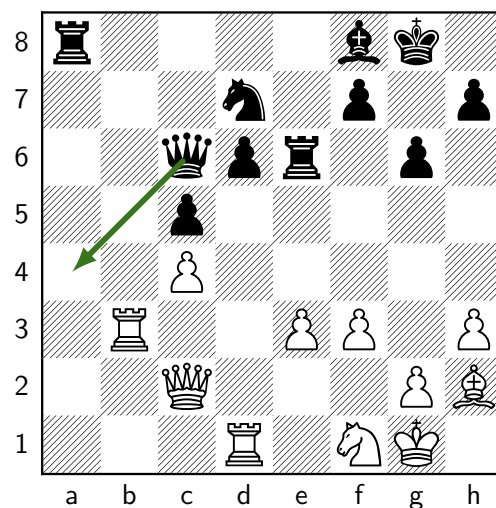
(a)



(b)

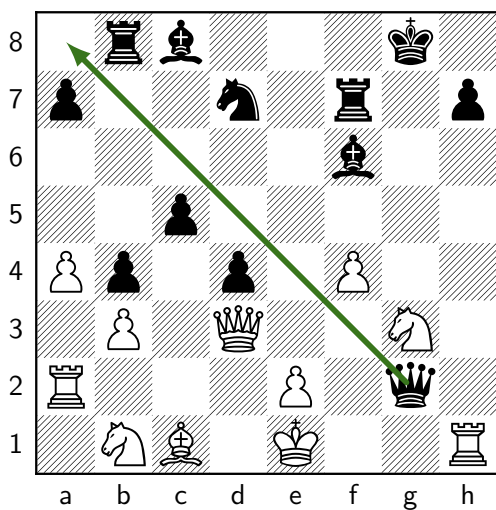


(c)

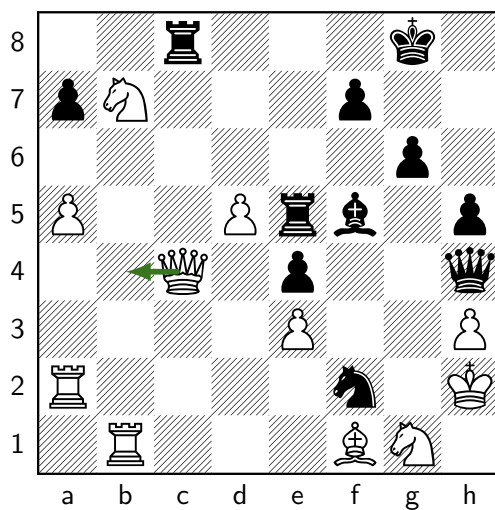


(d)

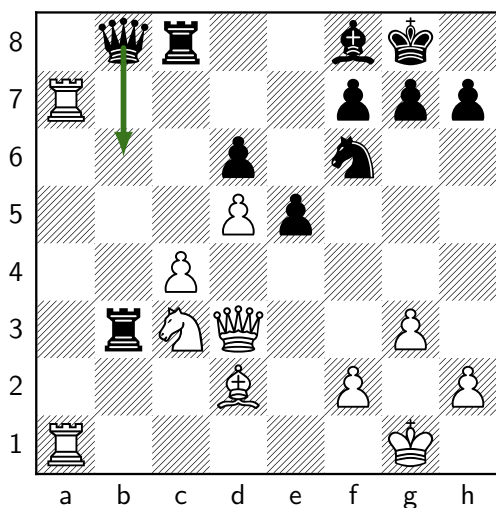
Figure 24: **Move 30, Black: Qg2.** (a) Our position; Melanie plays Qg2 (d5g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2604` (ply 74) [structural distance $d=82$, QAP obj=64]: the master played Qa5; under μ our Qd4 plays the role of that game’s Qb4; cross-type roles: our pf4 as their Re1, our qc2 as their Bg2; transferred move d5g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1548` (ply 48) [structural distance $d=80$, QAP obj=53]: the master played Qc7; under μ our Qd4 plays the role of that game’s Qg3; cross-type roles: our pf4 as their Ne1, our Qd5 as their Rc2; transferred move d5g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#221` (ply 59) [structural distance $d=87$, QAP obj=26]: the master played Qa4; under μ our Qd4 plays the role of that game’s Qc3; cross-type roles: our Nd7 as their Bf8, our pa4 as their Nd7; transferred move d5g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qg2 (summed QAP objective 167, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



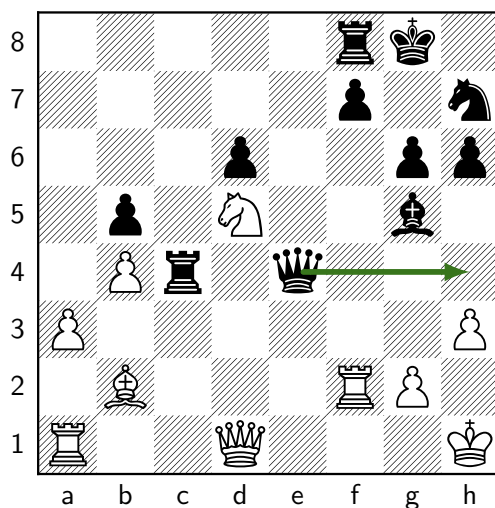
(a)



(b)

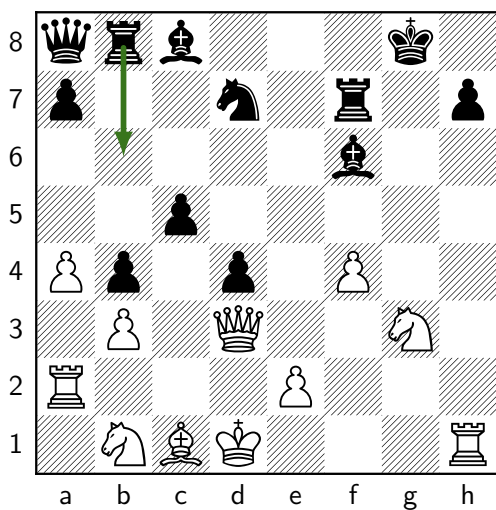


(c)

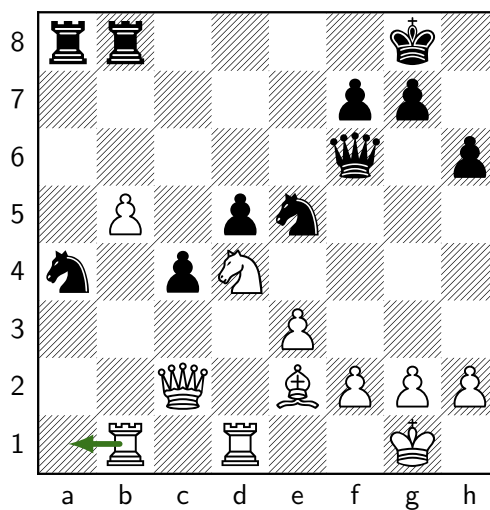


(d)

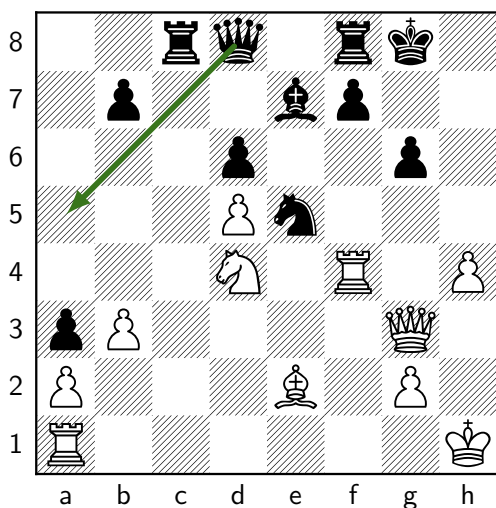
Figure 25: **Move 31, Black: Qa8**. (a) Our position; Melanie plays Qa8 (g2a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1476` (ply 60) [structural distance $d=107$, QAP obj=53]: the master played Qb4; under μ our Qg7 plays the role of that game’s Qc4; cross-type roles: our Rf7 as their Ng1, our Nd7 as their Ra2; transferred move g2a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1864` (ply 47) [structural distance $d=106$, QAP obj=45]: the master played Qb6; under μ our Qg7 plays the role of that game’s Qb1; cross-type roles: our pa4 as their Bf8, our bc1 as their Pg7; transferred move g2a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1845` (ply 57) [structural distance $d=106$, QAP obj=37]: the master played Qh4; under μ our Qg7 plays the role of that game’s Qe5; cross-type roles: our Pc5 as their Rf8, our Ph7 as their Nh7; transferred move g2a8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qa8 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



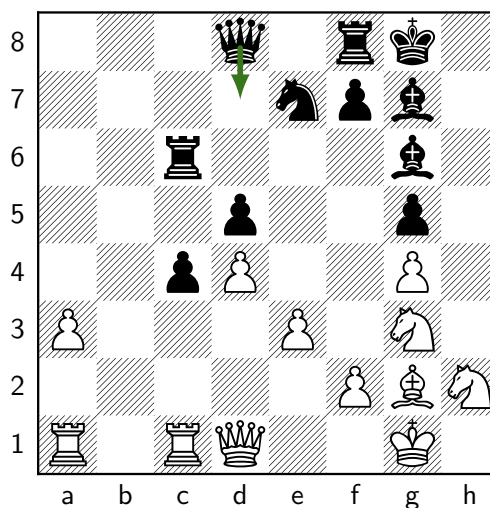
(a)



(b)

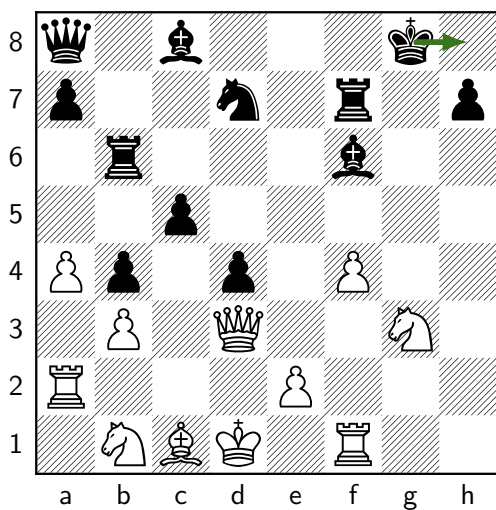


(c)

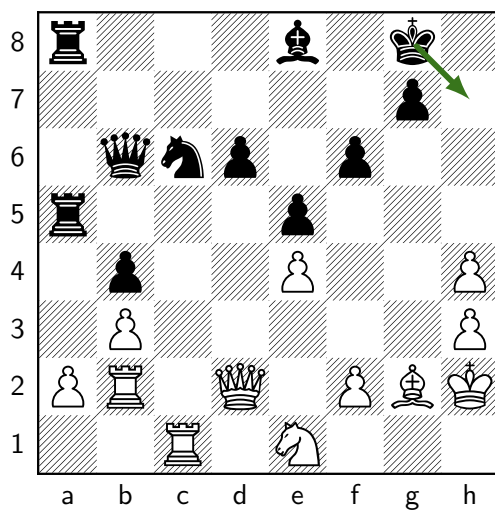


(d)

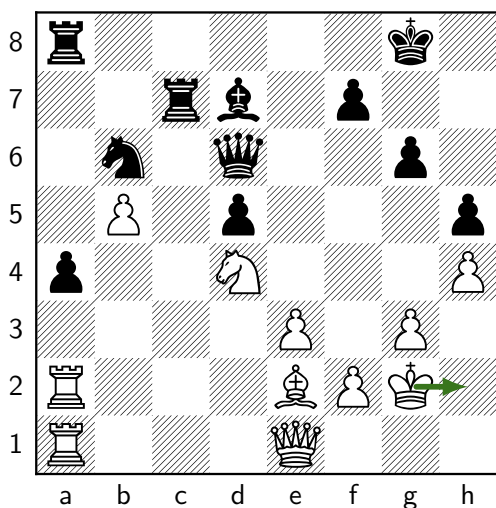
Figure 26: **Move 32, Black: Rb6.** (a) Our position; Melanie plays Rb6 (b8b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1950` (ply 52) [structural distance $d=97$, QAP obj=34]: the master played Ra1; under μ our Rb1 plays the role of that game’s Rb1; cross-type roles: our Nd7 as their Rd1, our Qa8 as their Be2; transferred move b8b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2935` (ply 55) [structural distance $d=98$, QAP obj=33]: the master played Qa5; under μ our Rb1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Rc8, our Pc5 as their Rf8; transferred move b8b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1535` (ply 53) [structural distance $d=96$, QAP obj=29]: the master played Qd7; under μ our Rb1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Rf8, our Bf6 as their Ne7; transferred move b8b6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rb6 (summed QAP objective 140, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



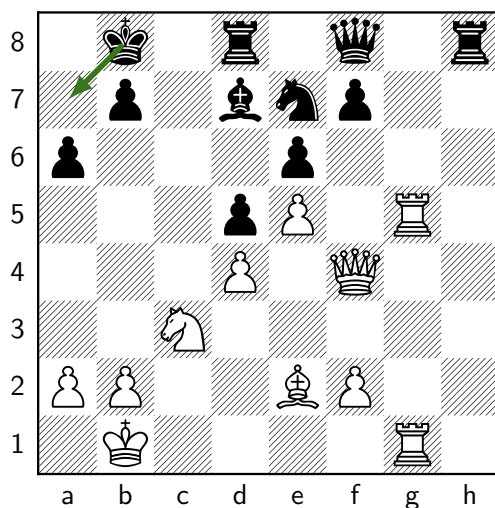
(a)



(b)

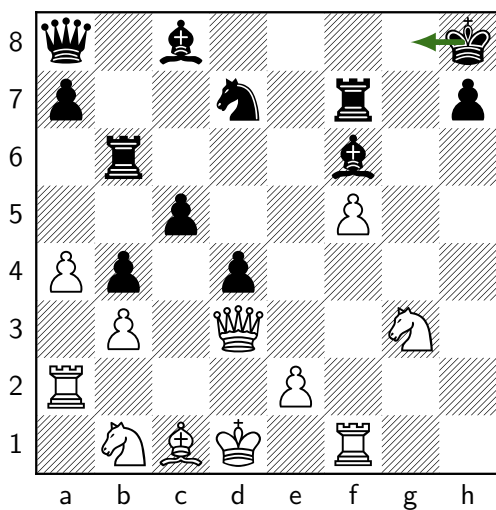


(c)

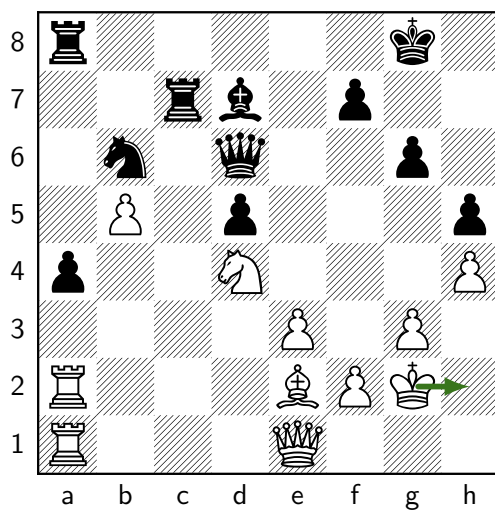


(d)

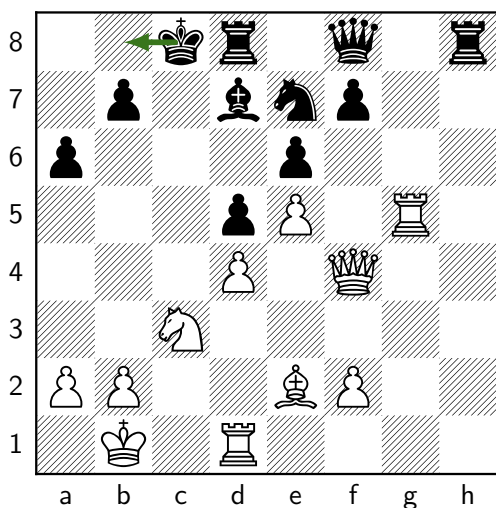
Figure 27: **Move 33, Black: Kh8 (g8h8)**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1629` (ply 63) [structural distance $d=105$, QAP obj=45]: the master played Kh7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Qa8* as their *Ra8*, our *Rf7* as their *Be8*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1443` (ply 68) [structural distance $d=96$, QAP obj=31]: the master played Kh2; under μ our *Kg1* plays the role of that game’s *Kg2*; cross-type roles: our *pe2* as their *Ra1*, our *Rf7* as their *Qe1*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2569` (ply 55) [structural distance $d=100$, QAP obj=28]: the master played Ka7; under μ our *Kg1* plays the role of that game’s *Kb1*; cross-type roles: our *Bf6* as their *Qf8*, our *Qa8* as their *Pb7*; transferred move g8h8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh8 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



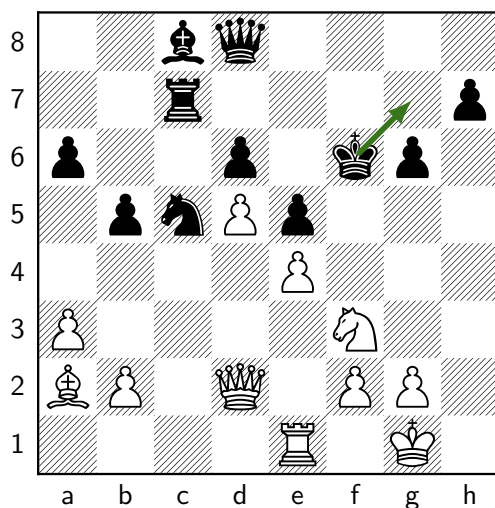
(a)



(b)

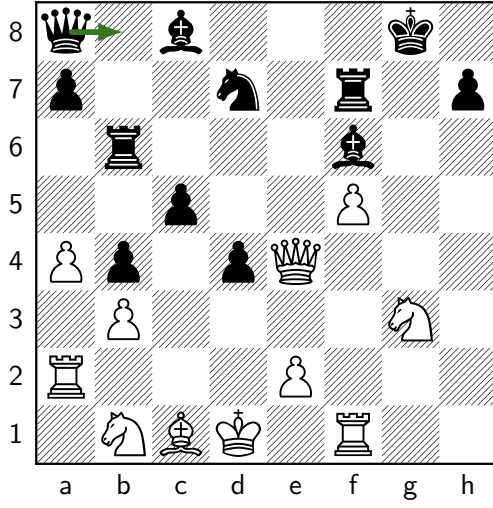


(c)

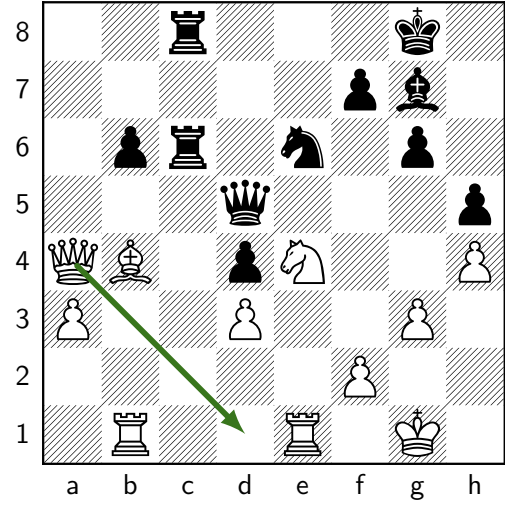


(d)

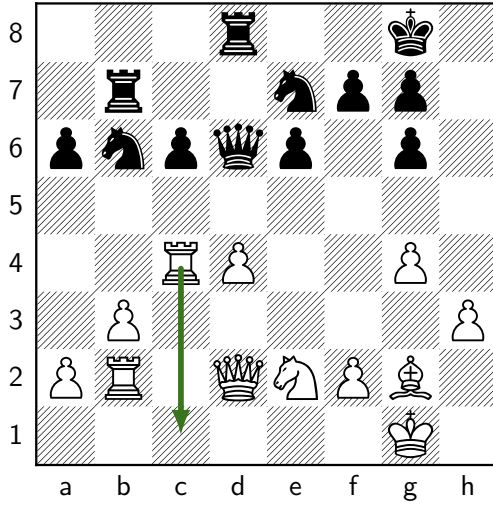
Figure 28: **Move 34, Black: Kg8.** (a) Our position; Melanie plays Kg8 (**h8g8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1443` (ply 68) [structural distance $d=116$, QAP obj=30]: the master played Kh2; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our pe2 as their Ra1, our Rf7 as their Qe1; transferred move **h8g8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#2569` (ply 53) [structural distance $d=120$, QAP obj=28]: the master played Kb8; under μ our Kh1 plays the role of that game’s Kc1; cross-type roles: our Ph7 as their Qf8, our Qa8 as their Pb7; transferred move **h8g8 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#1511` (ply 65) [structural distance $d=119$, QAP obj=19]: the master played Kg7; under μ our Kh1 plays the role of that game’s Kf3; cross-type roles: our nb1 as their Bc8, our Bf6 as their Qd8; transferred move **h8g8 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



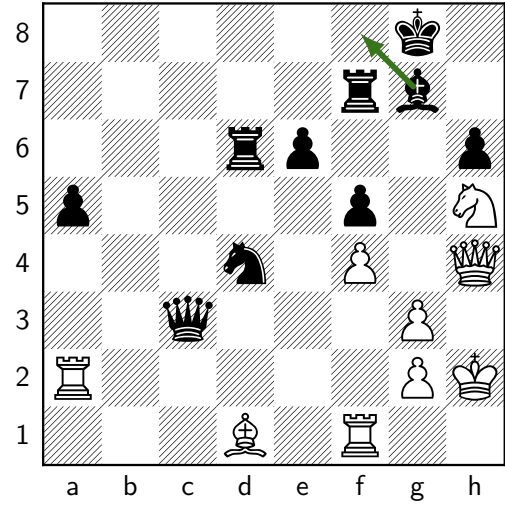
(a)



(b)

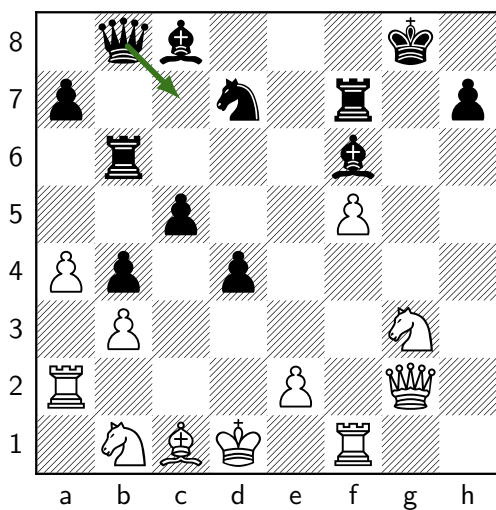


(c)

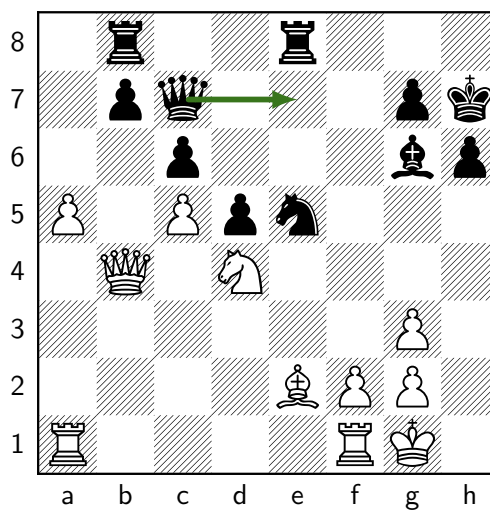


(d)

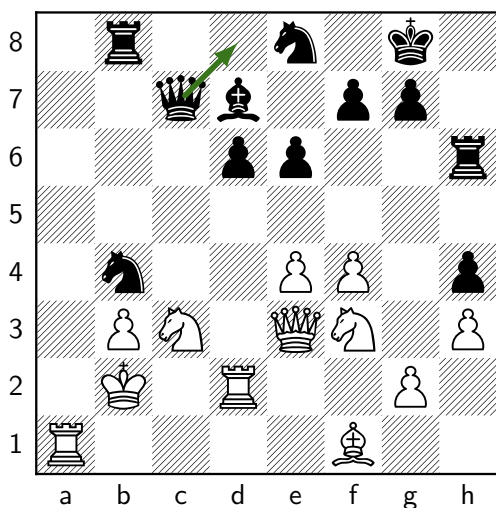
Figure 29: **Move 35, Black: Qb8**. (a) Our position; Melanie plays Qb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#719` (ply 66) [structural distance $d=99$, QAP obj=27]: the master played Qd1; under μ our Qa1 plays the role of that game’s Qa4; cross-type roles: our Bf6 as their Re1, our Bc8 as their Pf2; transferred move a8b8 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#1530` (ply 68) [structural distance $d=94$, QAP obj=21]: the master played Rc1; under μ our Qa1 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our pf5 as their Rb2, our Rb6 as their Qd2; transferred move a8b8 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#370` (ply 67) [structural distance $d=100$, QAP obj=19]: the master played Bf8; under μ our Qa1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Bf6 as their Rf7, our Nd7 as their Rd6; transferred move a8b8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qb8 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



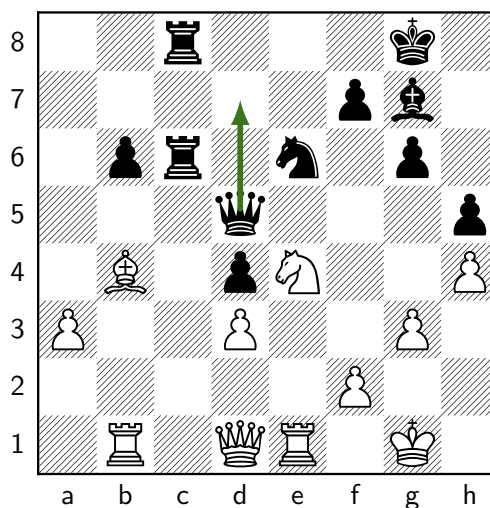
(a)



(b)

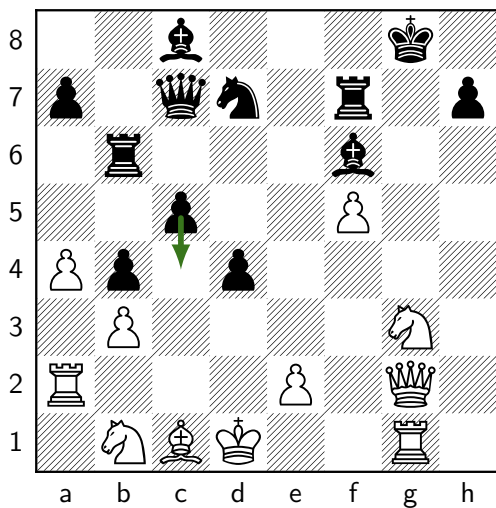


(c)

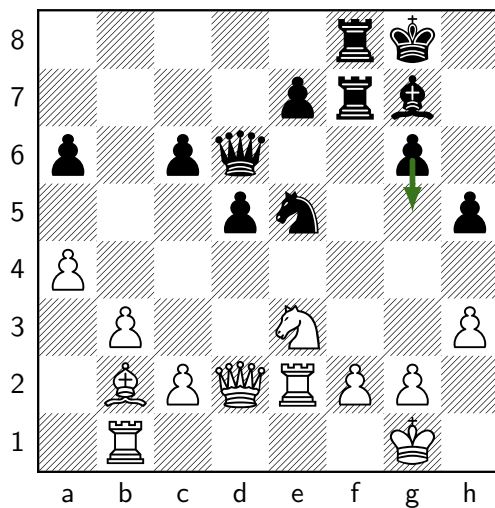


(d)

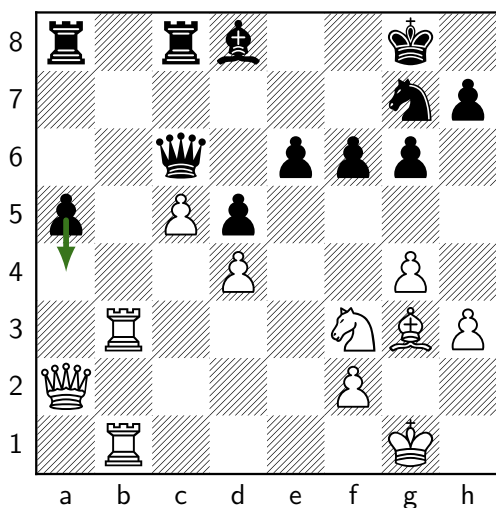
Figure 30: **Move 36, Black: Qc7.** (a) Our position; Melanie plays Qc7 (b8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#56` (ply 57) [structural distance $d=108$, QAP obj=29]: the master played Qe7; under μ our Qb1 plays the role of that game’s Qc2; cross-type roles: our Bc8 as their Pb7, our ng3 as their Pc6; transferred move b8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2606` (ply 57) [structural distance $d=113$, QAP obj=24]: the master played Qd8; under μ our Qb1 plays the role of that game’s Qc2; cross-type roles: our bc1 as their Rb8, our nb1 as their Bd7; transferred move b8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#719` (ply 67) [structural distance $d=111$, QAP obj=5]: the master played Qd7; under μ our Qb1 plays the role of that game’s Qd4; cross-type roles: our nb1 as their Pf7, our pb3 as their Rc6; transferred move b8c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc7 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



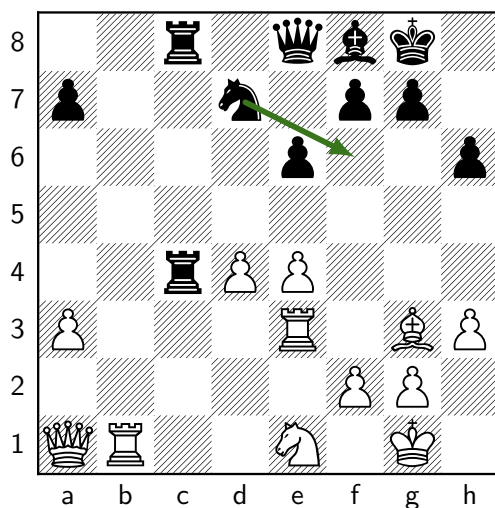
(a)



(b)

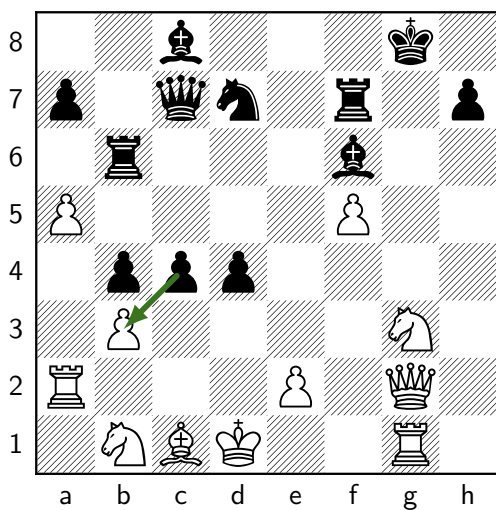


(c)

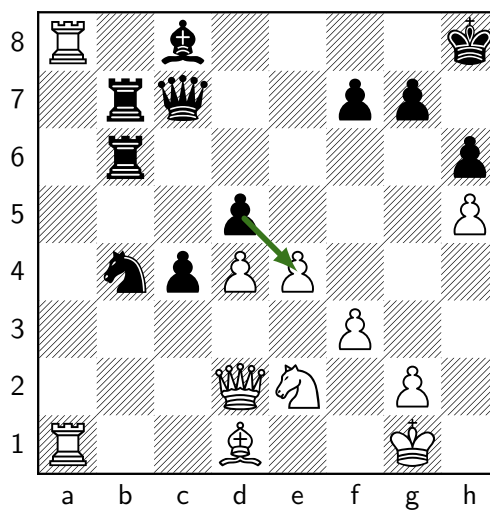


(d)

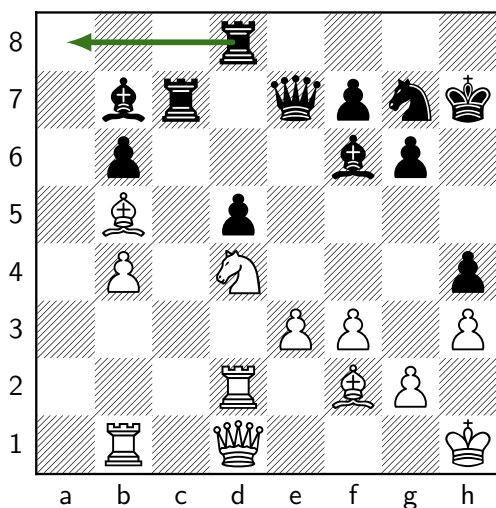
Figure 31: **Move 37, Black: c4.** (a) Our position; Melanie plays c4 (c5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2110` (ply 61) [structural distance $d=96$, QAP obj=19]: the master played g5; under μ our Pc_4 plays the role of that game’s Pg_3 ; cross-type roles: our qg_2 as their Pe_7 , our bc_1 as their Rf_7 ; transferred move c5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2949` (ply 63) [structural distance $d=110$, QAP obj=18]: the master played a4; under μ our Pc_4 plays the role of that game’s Pa_4 ; cross-type roles: our qg_2 as their Rc_8 , our pf_5 as their Bd_8 ; transferred move c5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2710` (ply 57) [structural distance $d=109$, QAP obj=17]: the master played Nf6; under μ our Pc_4 plays the role of that game’s Nd_2 (a **cross-type** correspondence); cross-type roles: our rg_1 as their Bf_8 , our Qc_7 as their Rc_4 ; transferred move c5c4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to c4 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



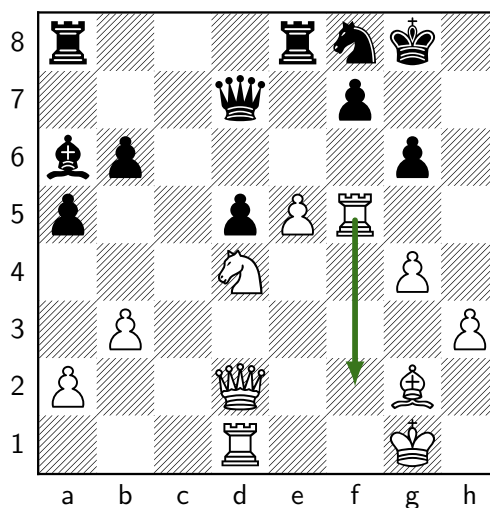
(a)



(b)

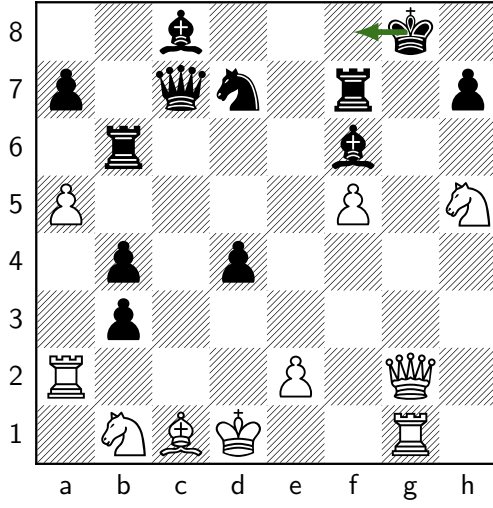


(c)

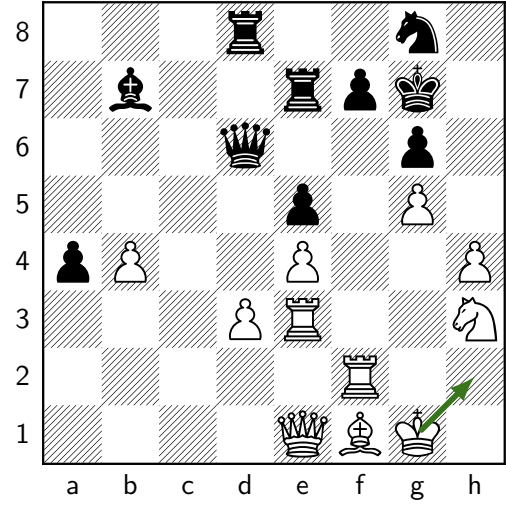


(d)

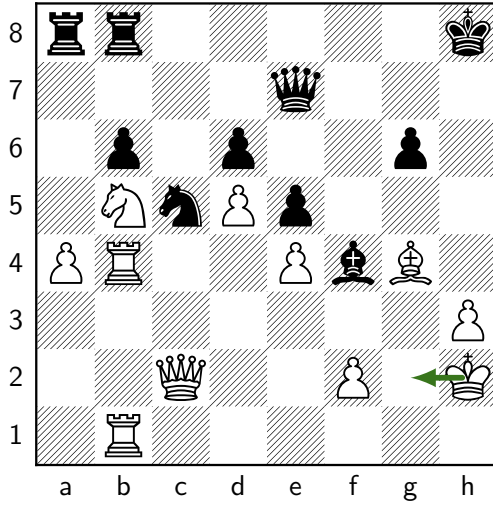
Figure 32: **Move 38, Black: cxb3.** (a) Our position; Melanie plays *cxb3* (*c4b3*). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from *ccr1_top_p0002#2307* (ply 59) [structural distance $d=104$, QAP obj=12]: the master played *dxe4*; under μ our *Pc5* plays the role of that game’s *Pd4*; cross-type roles: our *qg2* as their *ra8*, our *ng3* as their *Bc8*; transferred move *c4b3* (**endorses this move**). (c) Retrieved from *ccr1_top_p0000#1804* (ply 71) [structural distance $d=104$, QAP obj=11]: the master played *Ra8*; under μ our *Pc5* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Rf7* as their *Bb7*, our *Nd7* as their *Rc7*; transferred move *c4b3* (**endorses this move**). (d) Retrieved from *ccr1_top_p0003#1243* (ply 60) [structural distance $d=109$, QAP obj=32]: the master played *Rf2*; under μ our *Rb3* plays the role of that game’s *Rf5*; cross-type roles: our *Ph7* as their *Rd1*, our *nb1* as their *Pa2*; transferred move *b6b8*. Synthesis: 2 retrieved analogs transferred to *cxb3* (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



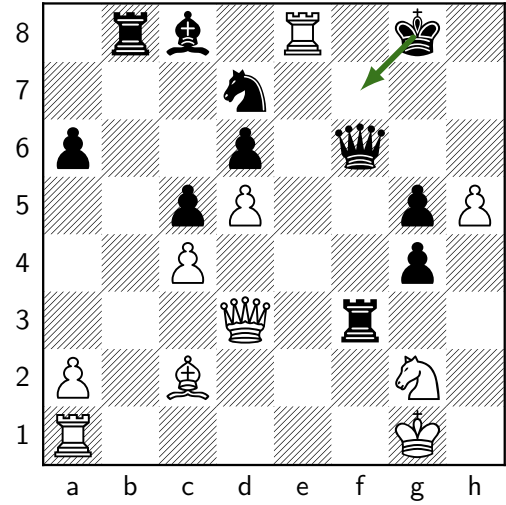
(a)



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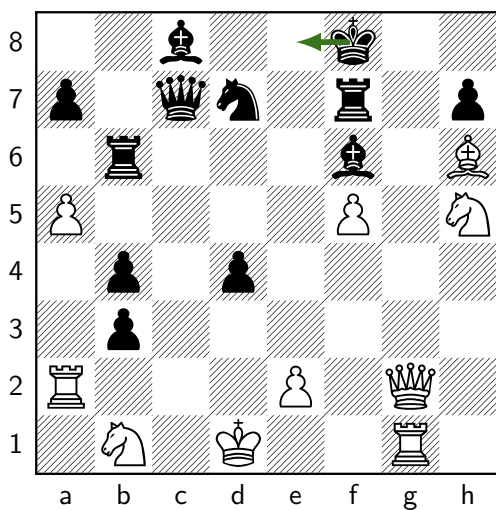


(c)

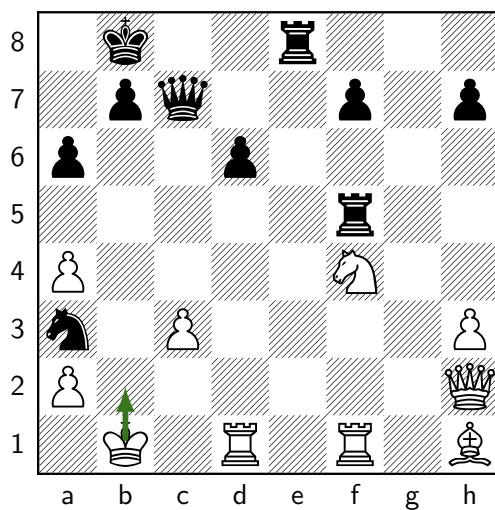


(d)

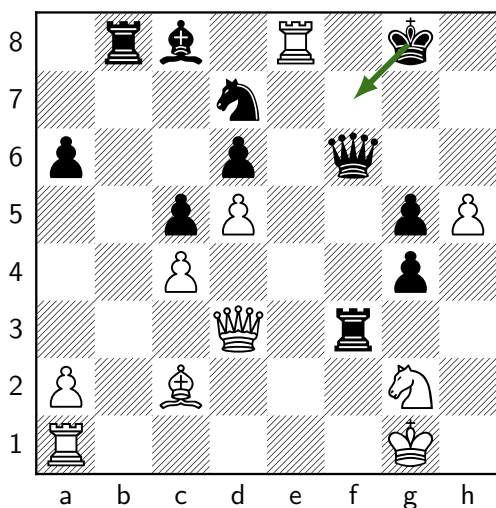
Figure 33: **Move 39, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1947` (ply 80) [structural distance $d=111$, QAP obj=5]: the master played Kh2; under μ our Kg1 plays the role of that game's Kg1; cross-type roles: our Rf7 as their Qe1, our Qc7 as their Rf2; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2042` (ply 70) [structural distance $d=128$, QAP obj=28]: the master played Kg2; under μ our Kg1 plays the role of that game's Kh2; cross-type roles: our Nd7 as their Rb1, our nb1 as their Qc2; transferred move g8h8. (d) Retrieved from `ccr1_top_p0002#1291` (ply 69) [structural distance $d=125$, QAP obj=27]: the master played Kf7; under μ our Kg1 plays the role of that game's Kg1; cross-type roles: our qg2 as their re8, our Rf7 as their Nd7; transferred move g8h8. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



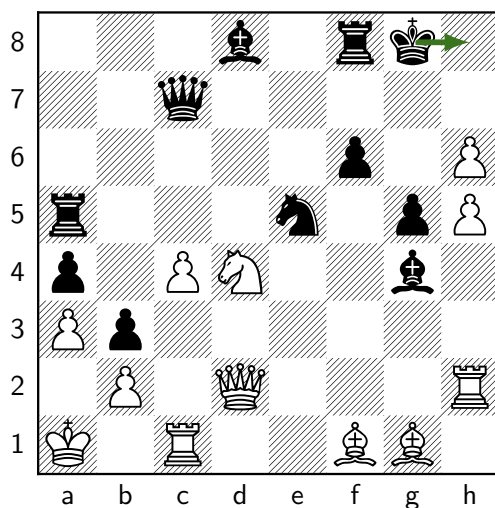
(a)



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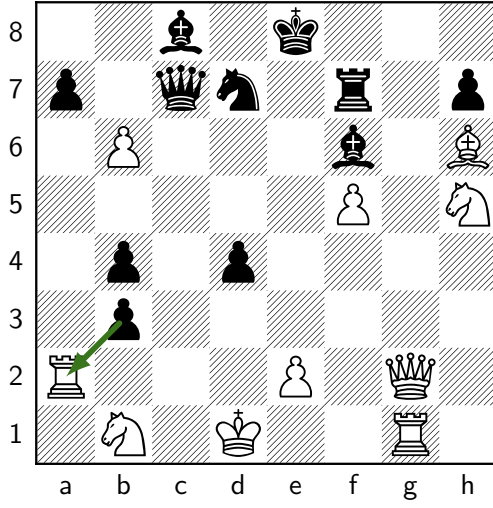


(c)

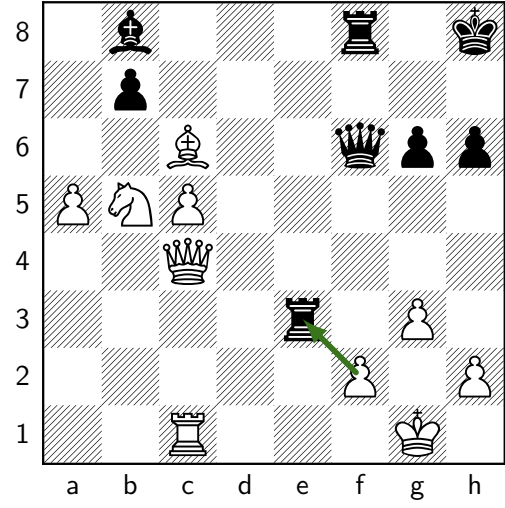


(d)

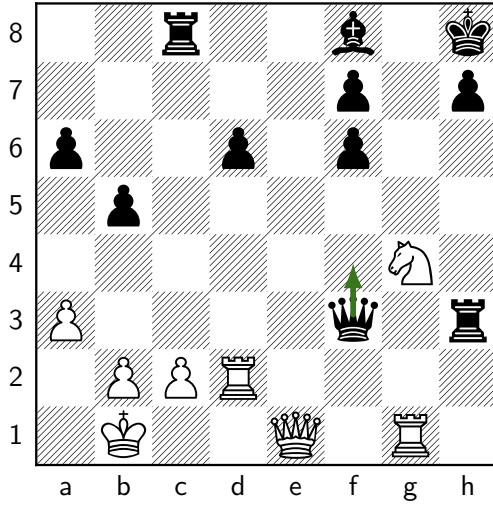
Figure 34: **Move 40, Black: Ke8.** (a) Our position; Melanie plays Ke8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1264` (ply 78) [structural distance $d=138$, QAP obj=72]: the master played Kb2; under μ our Kf1 plays the role of that game's Kb1; cross-type roles: our Nd7 as their Rf1, our pe2 as their Bh1; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1291` (ply 69) [structural distance $d=140$, QAP obj=69]: the master played Kf7; under μ our Kf1 plays the role of that game's Kg1; cross-type roles: our Qc7 as their Bc8, our bh6 as their re8; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#889` (ply 69) [structural distance $d=142$, QAP obj=5]: the master played Kh8; under μ our Kf1 plays the role of that game's Kg1; cross-type roles: our Rb6 as their Bd8, our Nd7 as their Rf8; transferred move f8e8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ke8 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



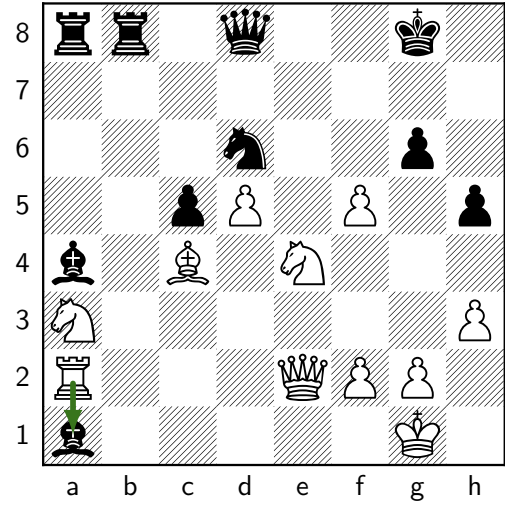
(a)



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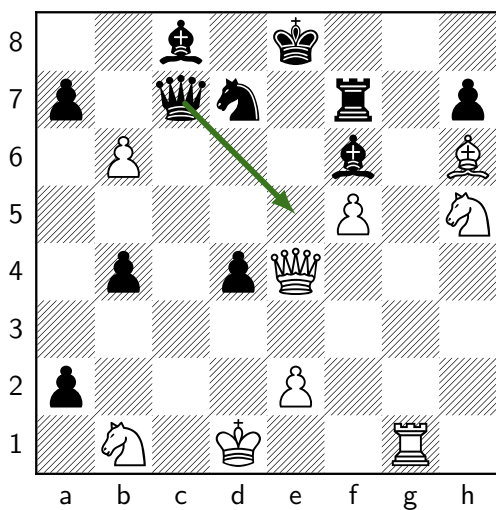


(c)

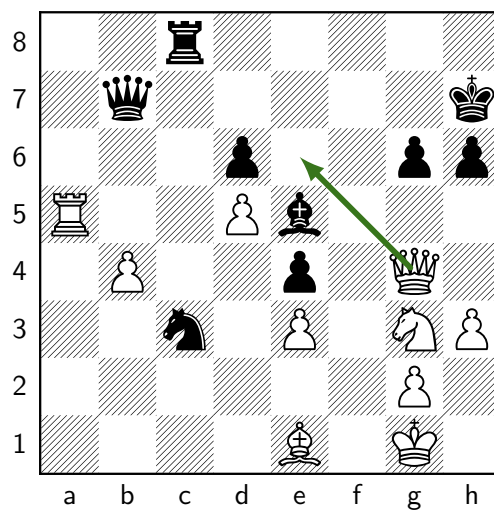


(d)

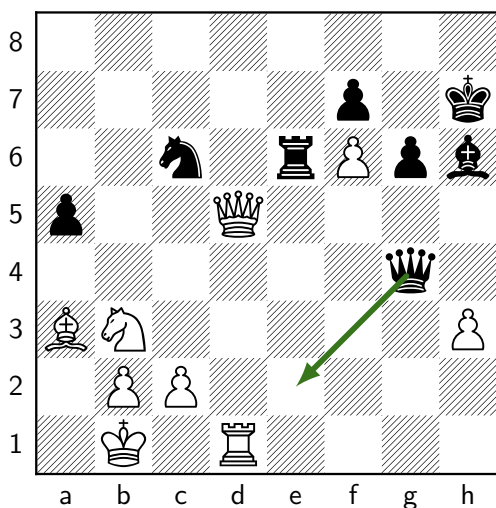
Figure 35: **Move 41, Black: bxa2.** (a) Our position; Melanie plays **bxa2** (**b3a2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#380` (ply 76) [structural distance $d=123$, QAP obj=24]: the master played **fxe3**; under μ our *Pb6* plays the role of that game’s *Pf2*; cross-type roles: our *Pb4* as their *Rc1*, our *bh6* as their *Pg3*; transferred move **b3a2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1303` (ply 67) [structural distance $d=121$, QAP obj=10]: the master played **Qf4**; under μ our *Pb6* plays the role of that game’s *Qf6* (a **cross-type** correspondence); cross-type roles: our *Qc7* as their *Rc8*, our *pf5* as their *Bf8*; transferred move **b3a2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#955` (ply 64) [structural distance $d=115$, QAP obj=10]: the master played **Rxa1**; under μ our *Pb6* plays the role of that game’s *Ra2* (a **cross-type** correspondence); cross-type roles: our *ra2* as their *ba1*, our *Rf7* as their *Qe2*; transferred move **b3a2** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **bxa2** (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



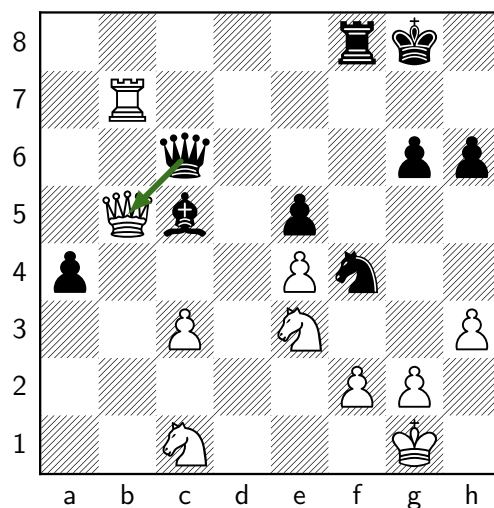
(a)



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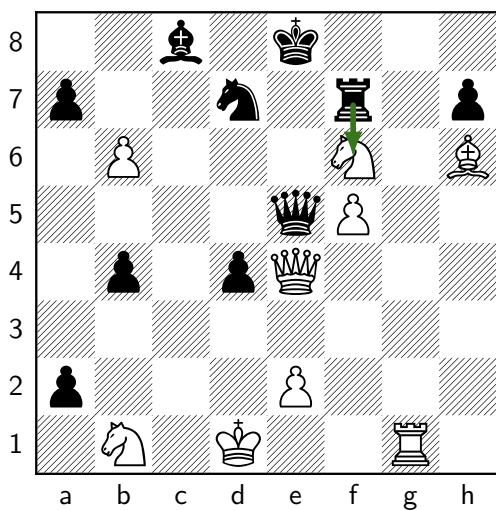


(c)

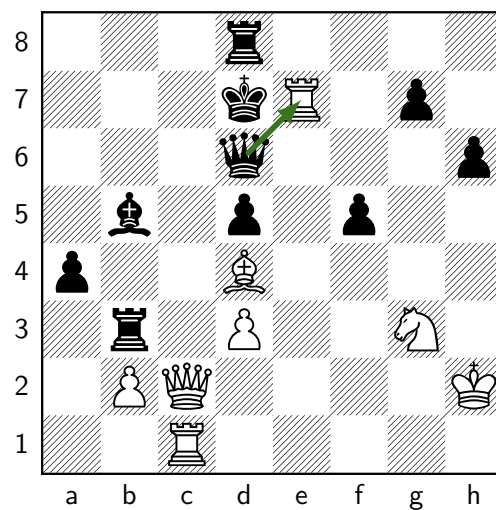


(d)

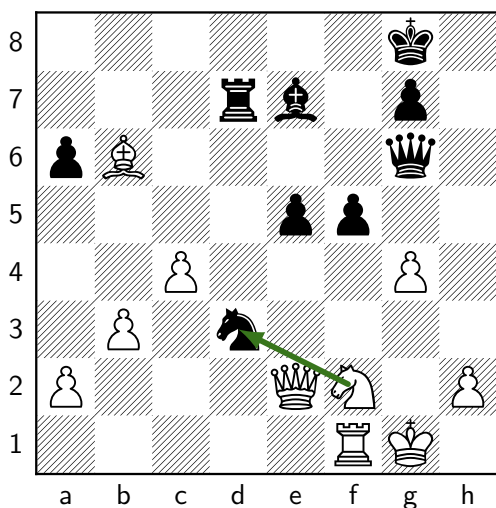
Figure 36: **Move 42, Black: Qe5**. (a) Our position; Melanie plays Qe5 (c7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3436` (ply 82) [structural distance $d=119$, QAP obj=15]: the master played Qe6; under μ our Qc2 plays the role of that game’s Qg4; cross-type roles: our Rf7 as their Be1, our Pd4 as their nc3; transferred move c7e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1248` (ply 87) [structural distance $d=119$, QAP obj=12]: the master played Qe2; under μ our Qc2 plays the role of that game’s Qg5; cross-type roles: our Rf7 as their Pf7, our Pd4 as their Nc6; transferred move c7e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#929` (ply 99) [structural distance $d=114$, QAP obj=12]: the master played Qxb5; under μ our Qc2 plays the role of that game’s Qc3; cross-type roles: our bh6 as their Rf8, our qe4 as their rb7; transferred move c7e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qe5 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



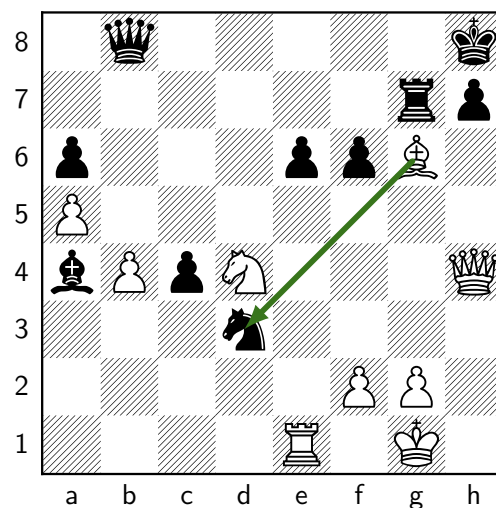
(a)



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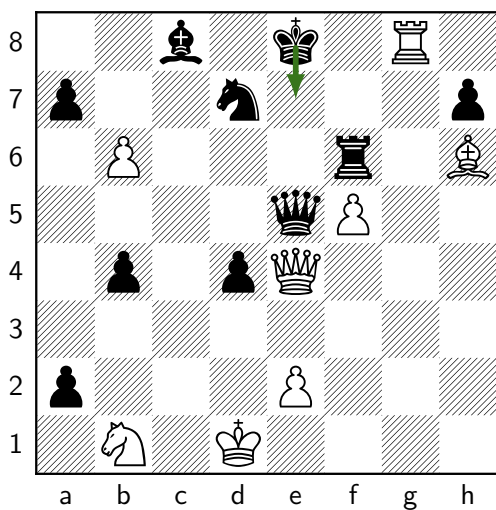


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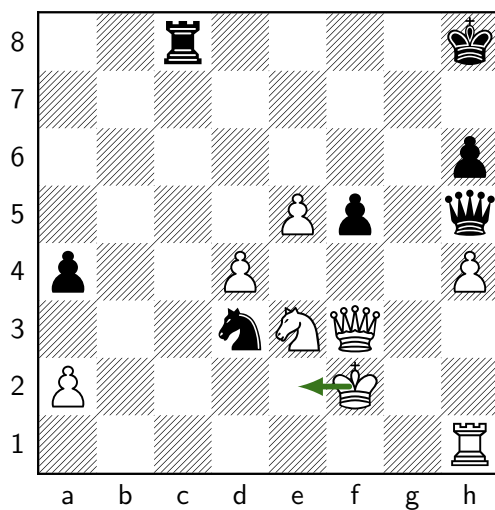


(d)

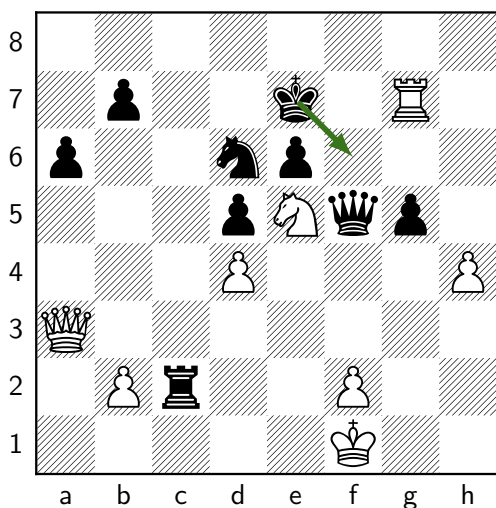
Figure 37: **Move 43, Black: Rxf6**. (a) Our position; Melanie plays Rxf6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1912` (ply 85) [structural distance $d=130$, QAP obj=44]: the master played Qxe7; under μ our Rf2 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Qe5 as their Rd8, our nf6 as their re7; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3289` (ply 74) [structural distance $d=120$, QAP obj=11]: the master played Nxd3; under μ our Rf2 plays the role of that game’s Nf2 (a **cross-type** correspondence); cross-type roles: our nb1 as their Pa2, our qe4 as their Pb3; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#850` (ply 70) [structural distance $d=122$, QAP obj=8]: the master played Bxd3; under μ our Rf2 plays the role of that game’s Bg6 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Re1, our bh6 as their Pf2; transferred move f7f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxf6 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



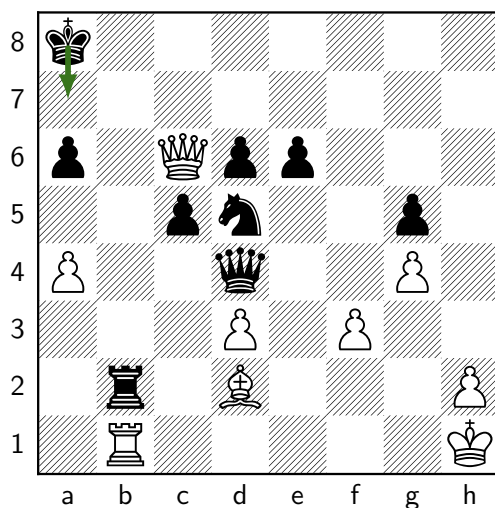
(a)



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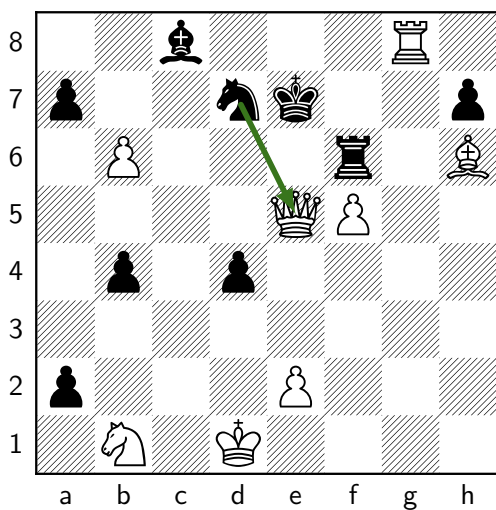


(c)

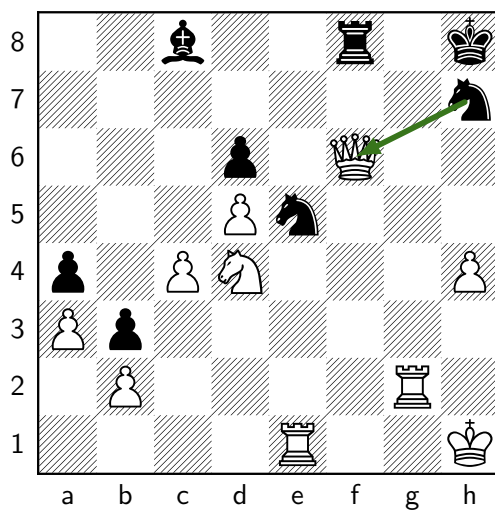


(d)

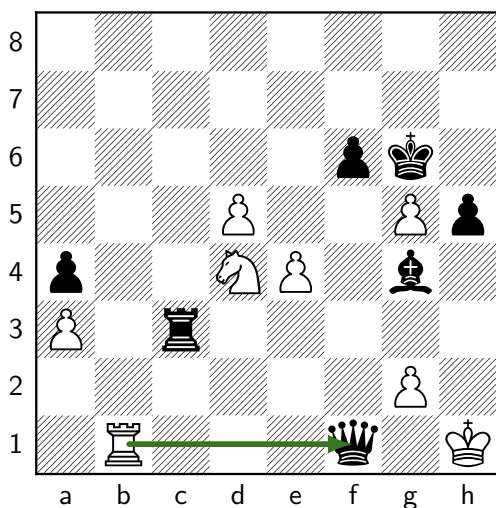
Figure 38: **Move 44, Black: Ke7.** (a) Our position; Melanie plays Ke7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#905` (ply 88) [structural distance $d=111$, QAP obj=86]: the master played Ke2; under μ our Ke1 plays the role of that game’s Kf2; cross-type roles: our Nd7 as their Rh1, our rg8 as their nd3; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#327` (ply 95) [structural distance $d=111$, QAP obj=72]: the master played Kf6; under μ our Ke1 plays the role of that game’s Ke2; cross-type roles: our Bc8 as their Pa6, our qe4 as their ne5; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#3598` (ply 75) [structural distance $d=102$, QAP obj=69]: the master played Ka7; under μ our Ke1 plays the role of that game’s Ka1; cross-type roles: our rg8 as their qc6, our Nd7 as their Qd4; transferred move e8e7 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Ke7 (summed QAP objective 666, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



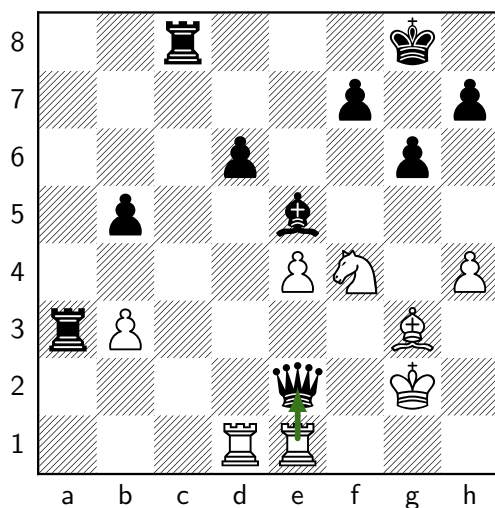
(a)



(b)

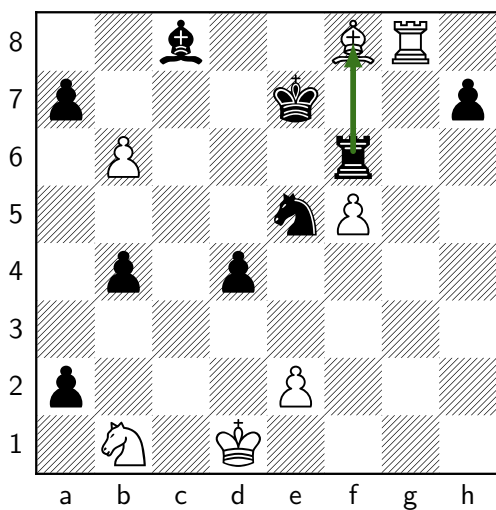


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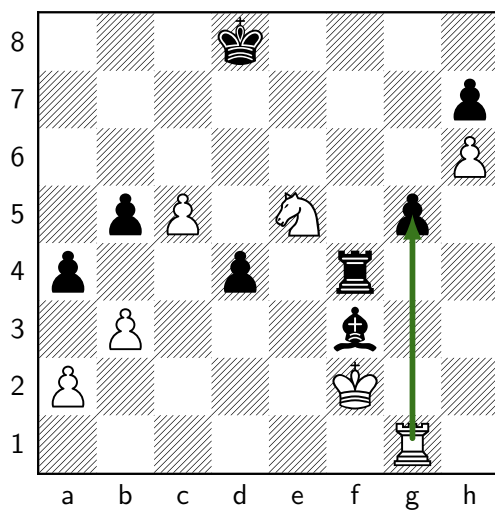


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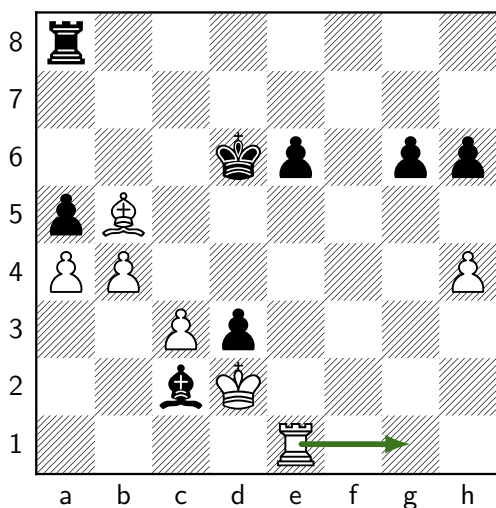
Figure 39: **Move 45, Black: Nxe5**. (a) Our position; Melanie plays Nxe5 (d7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2703` (ply 89) [structural distance $d=106$, QAP obj=76]: the master played Nxf6; under μ our Nd2 plays the role of that game’s Nh2; cross-type roles: our Pd4 as their Bc8, our nb1 as their pd5; transferred move d7e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2874` (ply 92) [structural distance $d=97$, QAP obj=71]: the master played Rxf1; under μ our Nd2 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Rf6 as their Pg2, our bh6 as their rc3; transferred move d7e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#660` (ply 76) [structural distance $d=110$, QAP obj=70]: the master played Rxe2; under μ our Nd2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our bh6 as their ra3, our rg8 as their Nf4; transferred move d7e5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Nxe5 (summed QAP objective 291, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



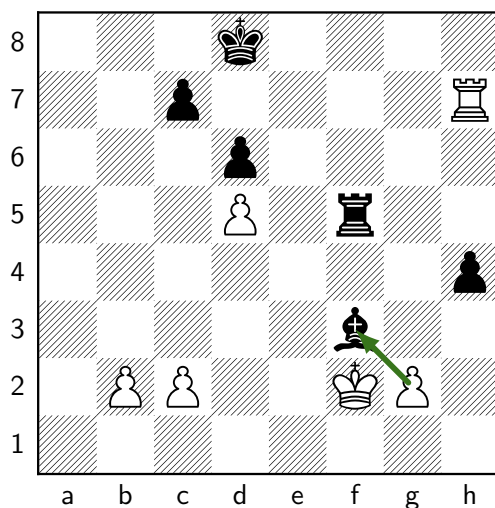
(a)



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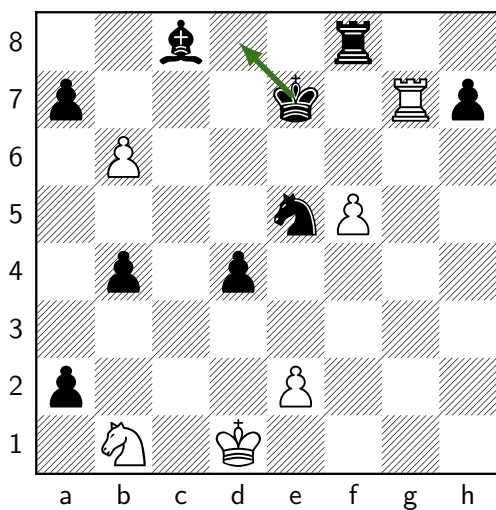


(c)

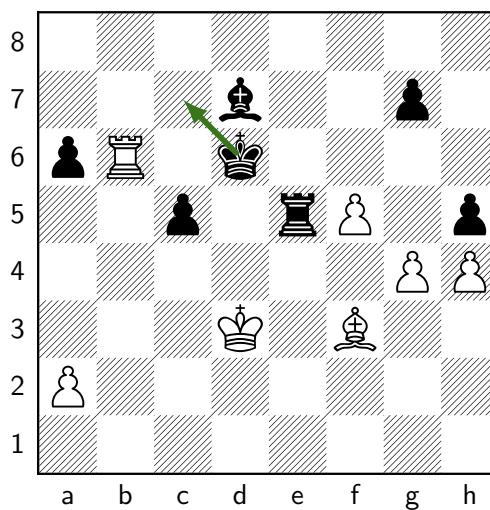


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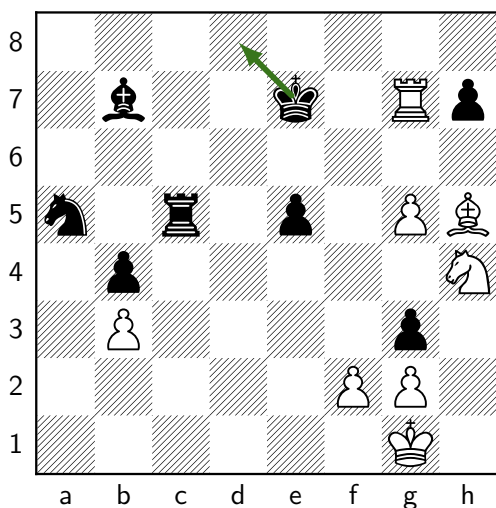
Figure 40: **Move 46, Black: Rxf8**. (a) Our position; Melanie plays Rxf8 (f6f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#2505` (ply 104) [structural distance $d=58$, QAP obj=7]: the master played Rxf5; under μ our Rf3 plays the role of that game’s Rg1; cross-type roles: our nb1 as their Pb3; transferred move f6f8 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#3941` (ply 104) [structural distance $d=66$, QAP obj=7]: the master played Rg1; under μ our Rf3 plays the role of that game’s Re1; cross-type roles: our rg8 as their pd3, our Ne5 as their Pb4; transferred move f6f8 (**endorses this move**). (d) Retrieved from `ccr1_top.p0001#2132` (ply 92) [structural distance $d=66$, QAP obj=5]: the master played gxf3; under μ our Rf3 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pc2, our nb1 as their pd6; transferred move f6f8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rxf8 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



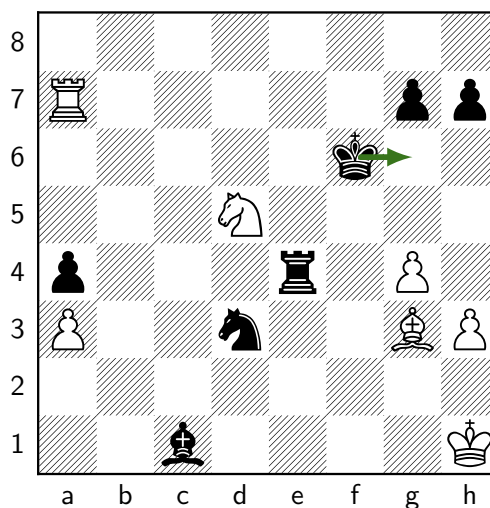
(a)



(b)



(c)



(d)

Figure 41: **Move 47, Black: Kd8.** (a) Our position; Melanie plays Kd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3772` (ply 85) [structural distance $d=42$, QAP obj=77]: the master played Kc7; under μ our Ke2 plays the role of that game’s Kd3; cross-type roles: our Ne5 as their ph4, our pb6 as their bf3; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2242` (ply 93) [structural distance $d=45$, QAP obj=71]: the master played Kd8; under μ our Ke2 plays the role of that game’s Ke2; cross-type roles: our Rf8 as their Bb7, our Bc8 as their Rc5; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2901` (ply 93) [structural distance $d=49$, QAP obj=69]: the master played Kg6; under μ our Ke2 plays the role of that game’s Kf3; cross-type roles: our rg7 as their nd5, our Bc8 as their Pa4; transferred move e7d8 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kd8 (summed QAP objective 551, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

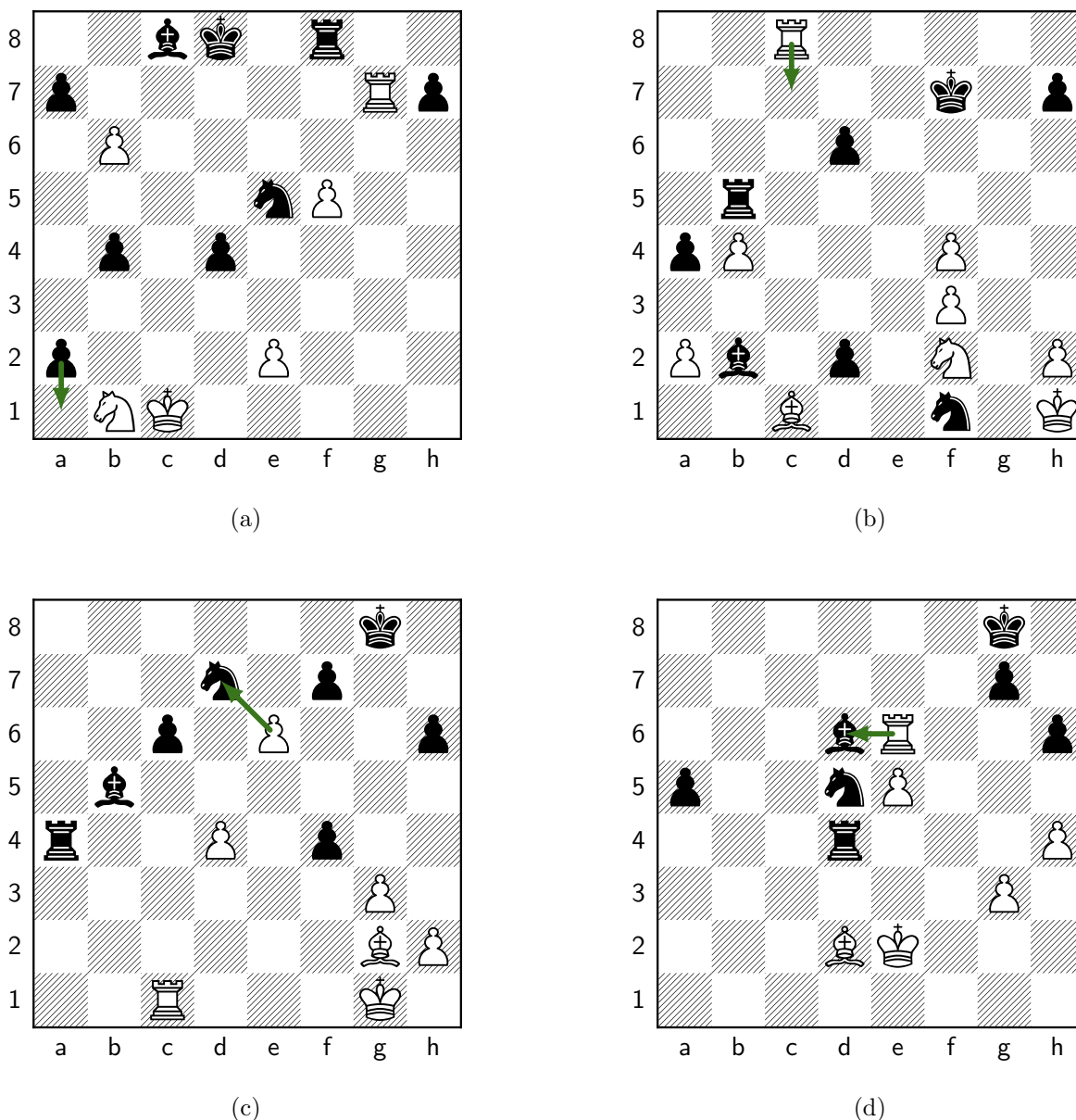
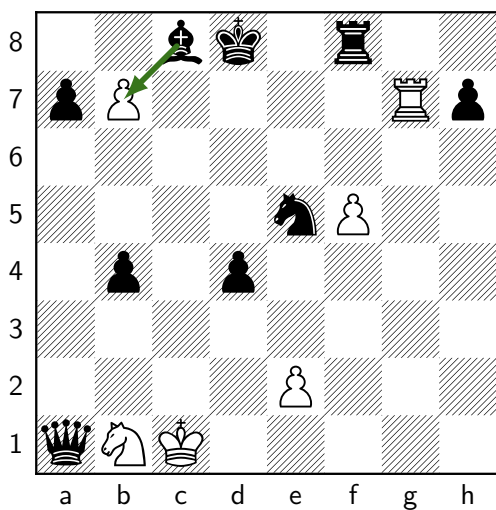
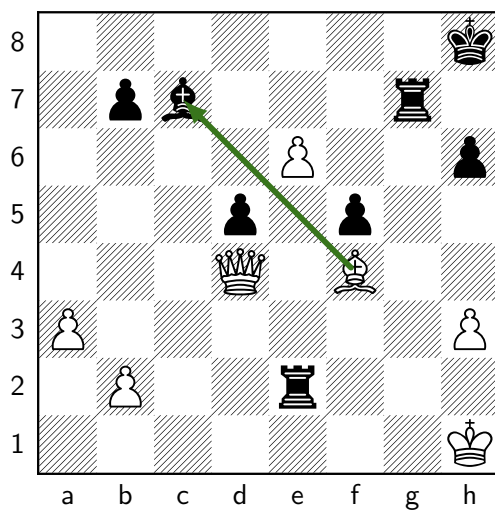


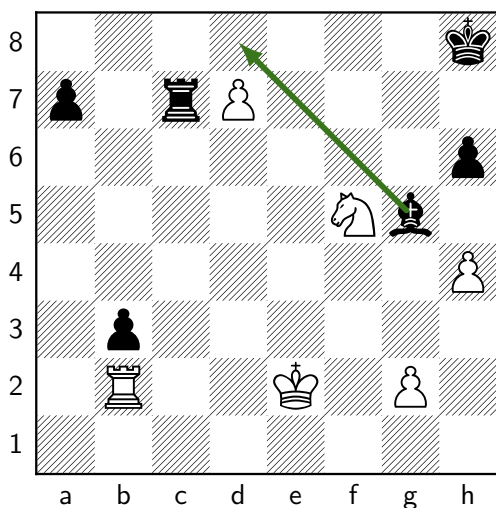
Figure 42: **Move 48, Black: a1=Q**. (a) Our position; Melanie plays a1=Q (a2a1q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1675` (ply 88) [structural distance $d=47$, QAP obj=41]: the master played Rc7+; under μ our Pa2 plays the role of that game’s Rc8 (a **cross-type** correspondence); cross-type roles: our nb1 as their Bc1, our Pd4 as their nf1; transferred move a7a5. (c) Retrieved from `ccr1_top_p0000#3468` (ply 82) [structural distance $d=45$, QAP obj=39]: the master played exd7; under μ our Pa7 plays the role of that game’s Pe6; cross-type roles: our pf5 as their bb5, our Ne5 as their ph6; transferred move a2a1b. (d) Retrieved from `ccr1_top_p0002#937` (ply 90) [structural distance $d=54$, QAP obj=39]: the master played Rxd6; under μ our Pa7 plays the role of that game’s Re6 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Bd2, our Bc8 as their Pg3; transferred move a2a1b. Synthesis: no retrieved analog transferred to a1=Q – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



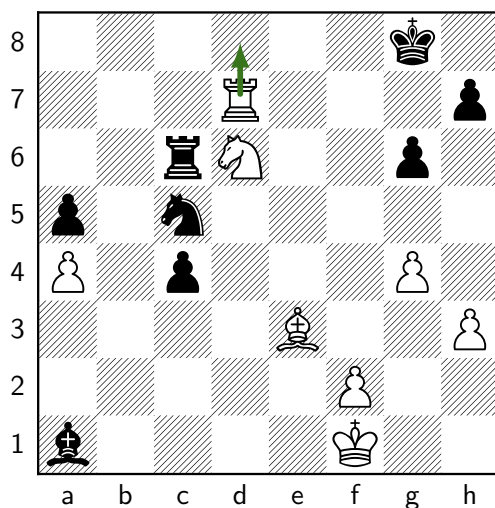
(a)



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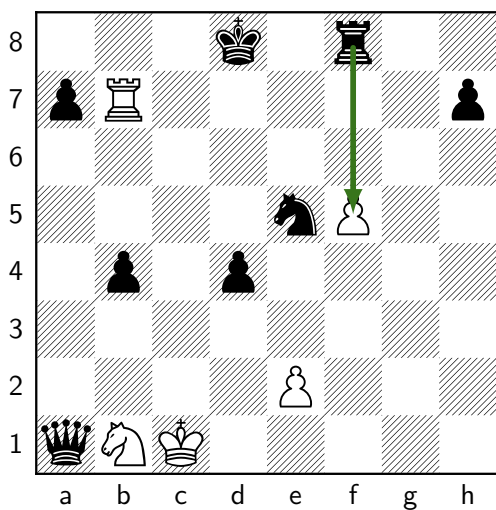


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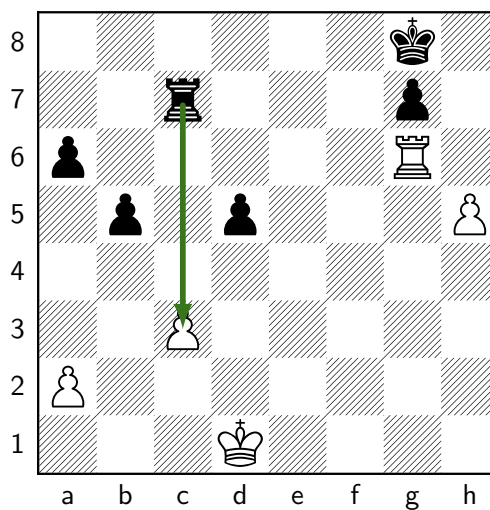


(d)

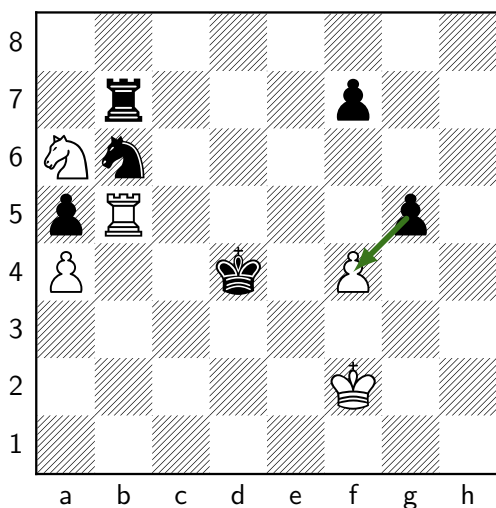
Figure 43: **Move 49, Black: Bxb7**. (a) Our position; Melanie plays Bxb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3673 (ply 100) [structural distance $d=54$, QAP obj=44]: the master played Bxc7; under μ our Bc1 plays the role of that game’s Bf4; cross-type roles: our nb1 as their pd5, our Rf8 as their ph6; transferred move c8b7 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#619 (ply 99) [structural distance $d=54$, QAP obj=37]: the master played Bd8; under μ our Bc1 plays the role of that game’s Bg4; cross-type roles: our Qa1 as their Rc7, our nb1 as their pd7; transferred move c8b7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1172 (ply 82) [structural distance $d=51$, QAP obj=37]: the master played Rd8+; under μ our Bc1 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Pb4 as their ba1, our rg7 as their Pf2; transferred move c8b7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bxb7 (summed QAP objective 265, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



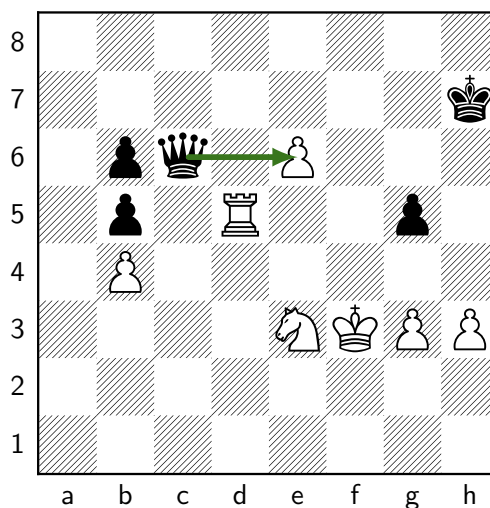
(a)



(b)

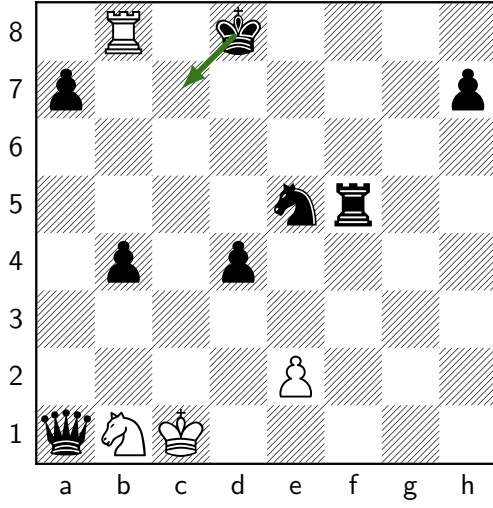


(c)

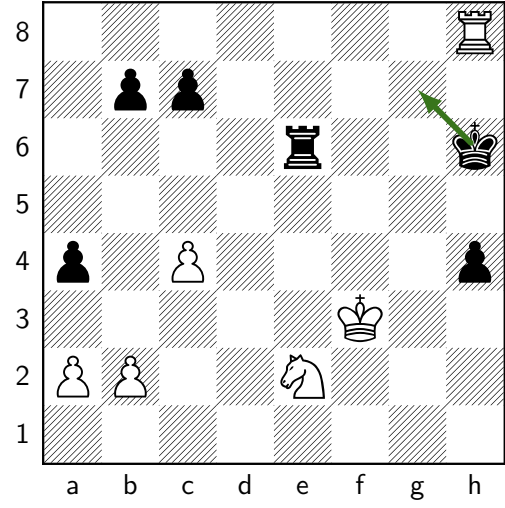


(d)

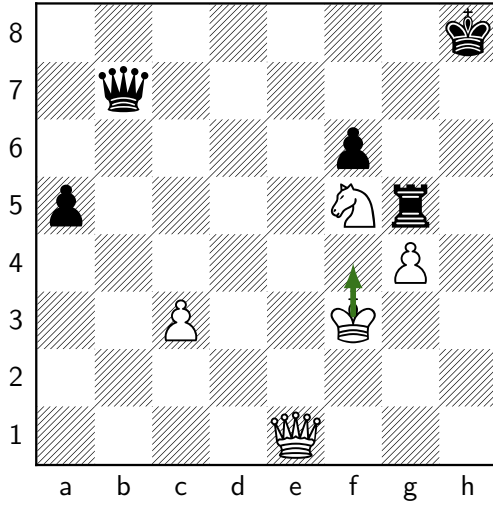
Figure 44: **Move 50, Black: Rxf5.** (a) Our position; Melanie plays Rxf5 (f8f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#689` (ply 83) [structural distance $d=47$, QAP obj=15]: the master played Rxc3; under μ our Rf1 plays the role of that game’s Rc2; cross-type roles: our nb1 as their Pg7, our Qa1 as their rg6; transferred move f8f5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3874` (ply 97) [structural distance $d=45$, QAP obj=48]: the master played gxf4; under μ our Pa2 plays the role of that game’s Pg4; cross-type roles: our Ne5 as their Rb7, our Rf8 as their Pf7; transferred move a7a6. (d) Retrieved from `ccrl_top_p0000#815` (ply 95) [structural distance $d=47$, QAP obj=31]: the master played Qxe6; under μ our Qa8 plays the role of that game’s Qc3; cross-type roles: our rb7 as their pe6, our nb1 as their rd5; transferred move a1a6. Synthesis: 1 retrieved analog transferred to Rxf5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



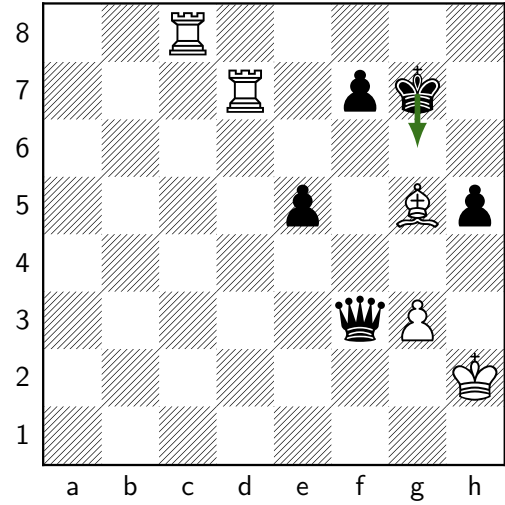
(a)



(b)

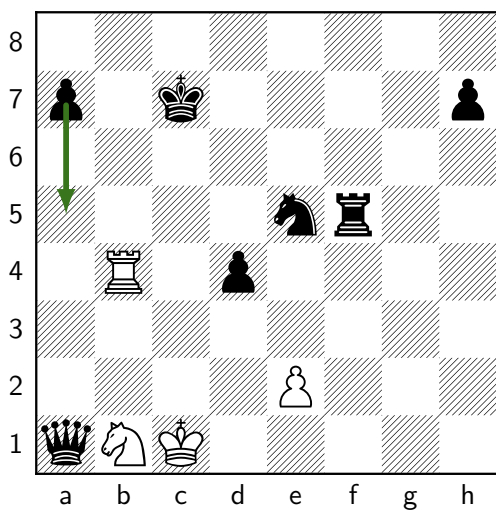


(c)

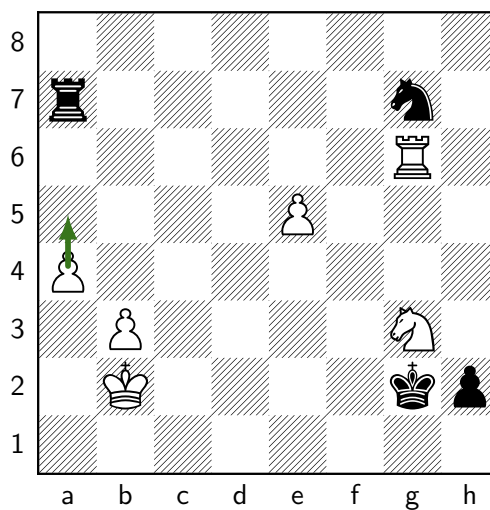


(d)

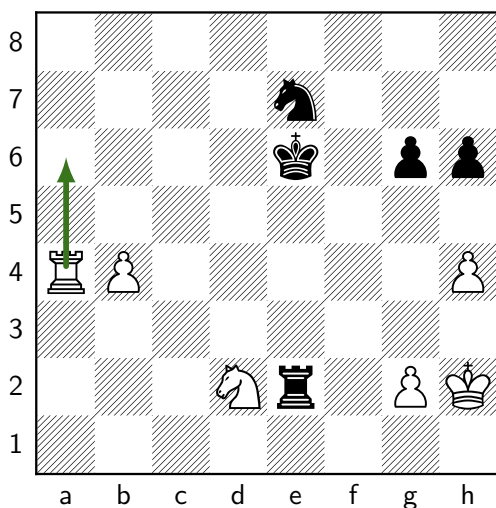
Figure 45: **Move 51, Black: Kc7**. (a) Our position; Melanie plays Kc7 (d8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3761` (ply 87) [structural distance $d=39$, QAP obj=68]: the master played Kg7; under μ our Kd1 plays the role of that game’s Kh3; cross-type roles: our Qa1 as their Re6, our nb1 as their Pa4; transferred move d8c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3456` (ply 116) [structural distance $d=38$, QAP obj=34]: the master played Kf4; under μ our Kd1 plays the role of that game’s Kf3; cross-type roles: our nb1 as their pa5, our Pd4 as their Nf5; transferred move d8c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3505` (ply 85) [structural distance $d=38$, QAP obj=6]: the master played Kg6; under μ our Kd1 plays the role of that game’s Kg2; cross-type roles: our Pa7 as their rd7, our Ne5 as their bg5; transferred move d8c7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kc7 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



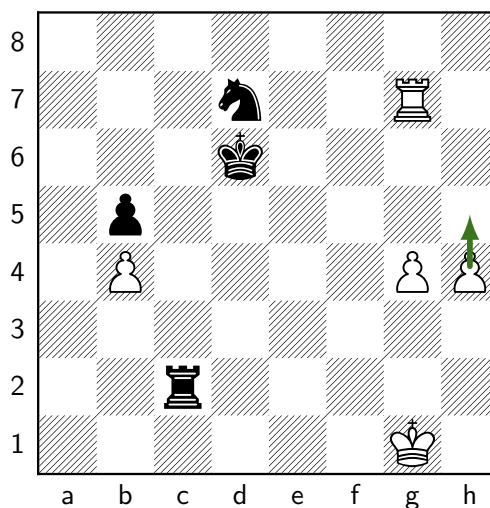
(a)



(b)

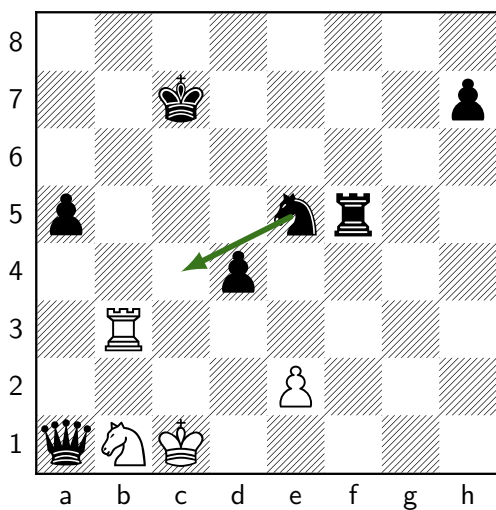


(c)

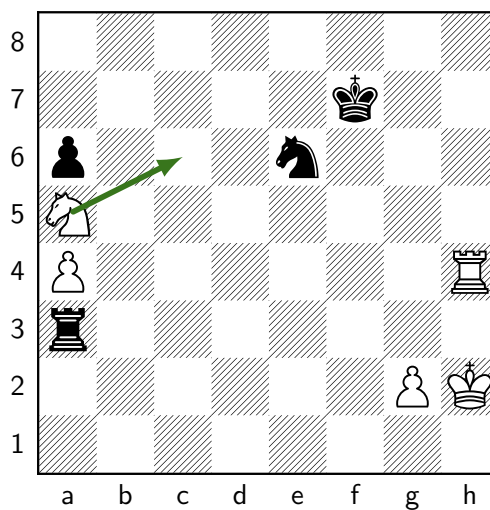


(d)

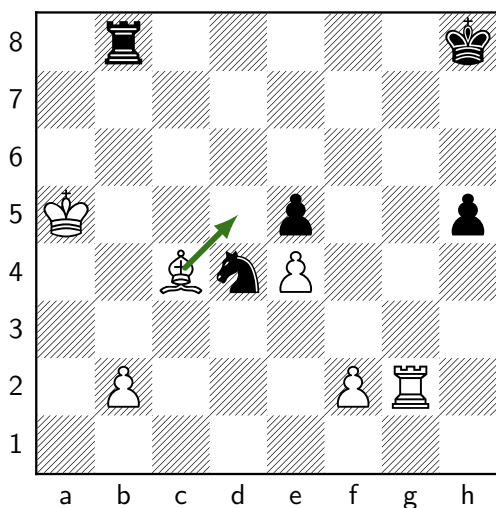
Figure 46: **Move 52, Black: a5.** (a) Our position; Melanie plays a5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3917` (ply 106) [structural distance $d=29$, QAP obj=22]: the master played a5; under μ our $Pa2$ plays the role of that game’s $Pa4$; cross-type roles: our $Ne5$ as their $Pb3$, our $Pd4$ as their $Ng3$; transferred move a7a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#326` (ply 90) [structural distance $d=36$, QAP obj=19]: the master played Ra6+; under μ our $Pa2$ plays the role of that game’s $Ra4$ (a **cross-type** correspondence); cross-type roles: our $Qa1$ as their $re2$, our $rb4$ as their $Pg2$; transferred move a7a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1662` (ply 96) [structural distance $d=35$, QAP obj=10]: the master played h5; under μ our $Pa2$ plays the role of that game’s $Ph4$; cross-type roles: our $Ne5$ as their $pb5$, our $Qa1$ as their $Rg7$; transferred move a7a5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a5 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



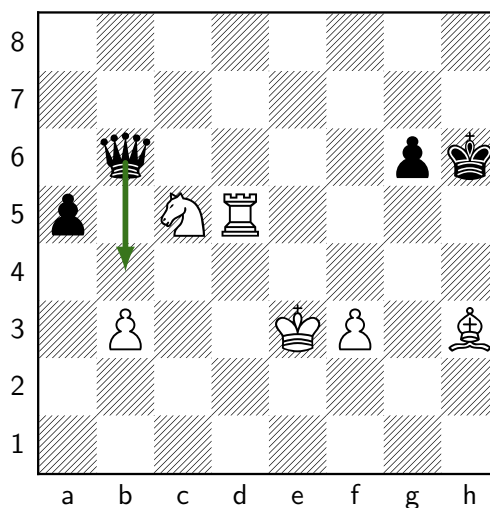
(a)



(b)

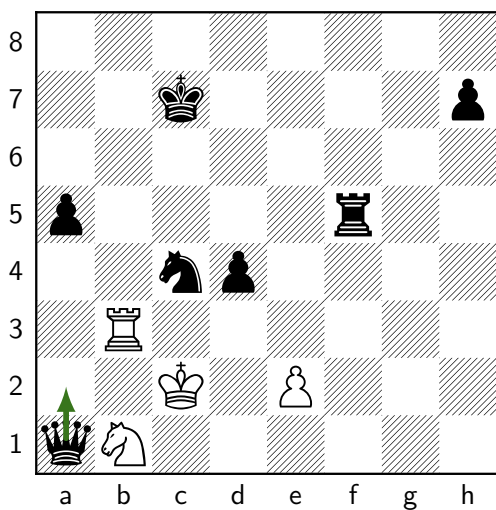


(c)

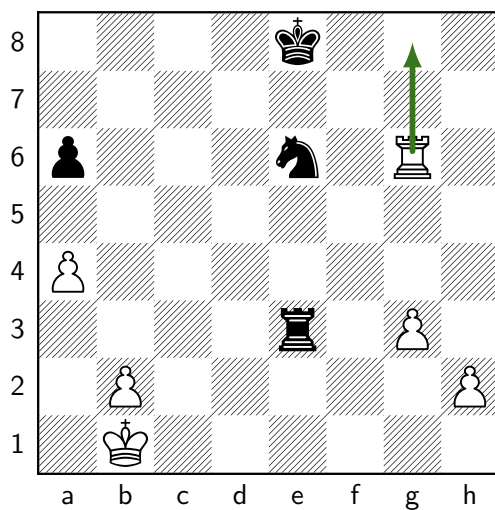


(d)

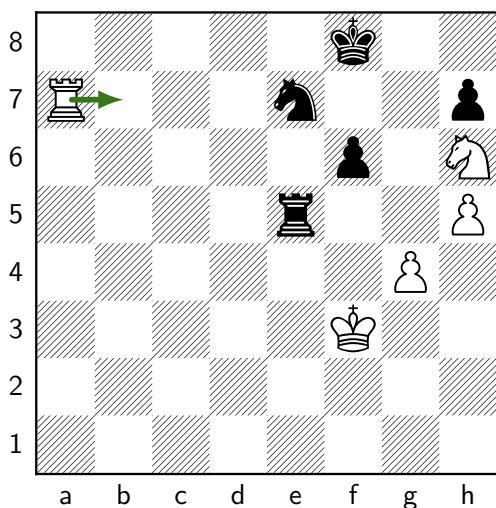
Figure 47: **Move 53, Black: Nc4.** (a) Our position; Melanie plays Nc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#996` (ply 116) [structural distance $d=38$, QAP obj=14]: the master played Nc6; under μ our Ne4 plays the role of that game’s Na5; cross-type roles: our Rf5 as their Pg2, our Qa1 as their ra3; transferred move e5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1485` (ply 100) [structural distance $d=38$, QAP obj=2]: the master played Bd5; under μ our Ne4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Qa1 as their Rg2, our pe2 as their nd4; transferred move e5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2779` (ply 111) [structural distance $d=32$, QAP obj=50]: the master played Qb4; under μ our Qa8 plays the role of that game’s Qb3; cross-type roles: our Ne5 as their bh3; transferred move a1a2. Synthesis: 2 retrieved analogs transferred to Nc4 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



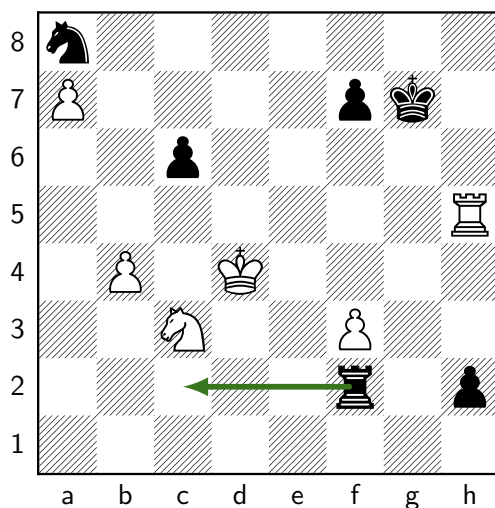
(a)



(b)

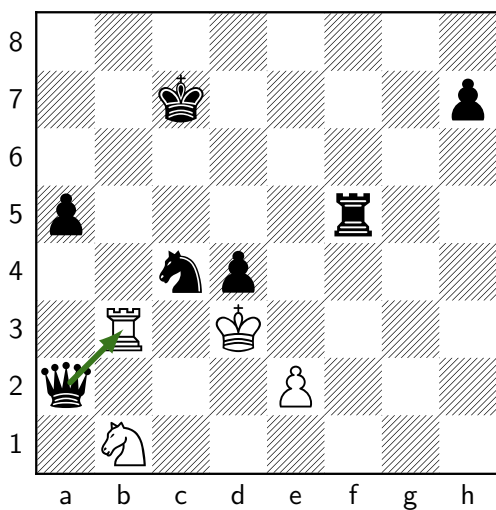


(c)

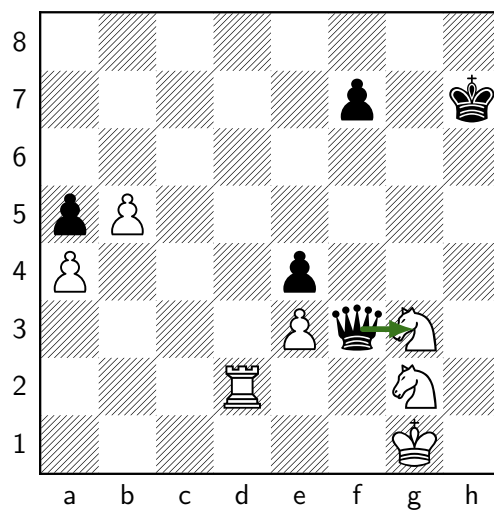


(d)

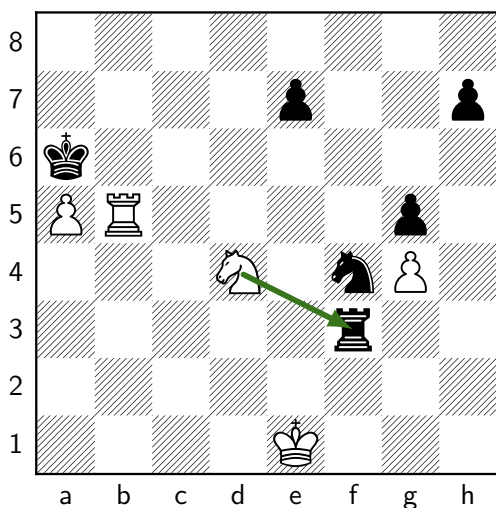
Figure 48: **Move 54, Black: Qa2+.** (a) Our position; Melanie plays Qa2+ (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1898` (ply 92) [structural distance $d=47$, QAP obj=12]: the master played Rg8+; under μ our Qa8 plays the role of that game’s Rg6 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Pb2, our Rf5 as their pa6; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2970` (ply 112) [structural distance $d=41$, QAP obj=9]: the master played Rb7; under μ our Qa8 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our Rf5 as their ph7; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2859` (ply 105) [structural distance $d=48$, QAP obj=9]: the master played Rc2; under μ our Qa8 plays the role of that game’s Rf7 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Na8, our Rf5 as their pa7; transferred move a1a2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qa2+ (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



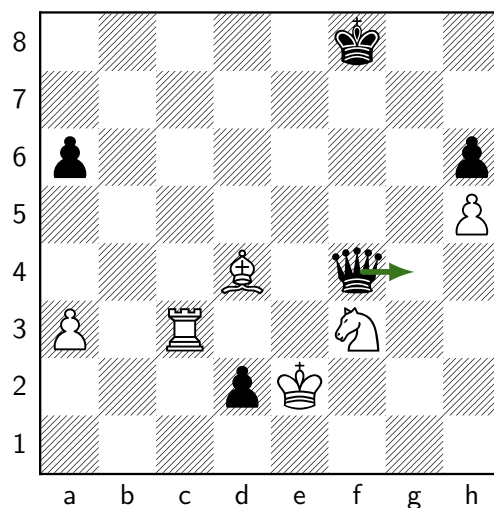
(a)



(b)

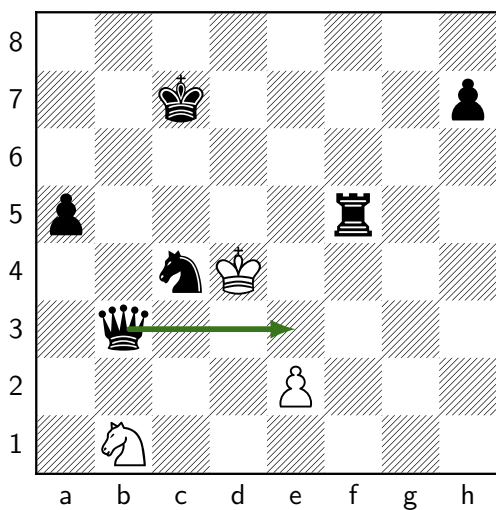


(c)

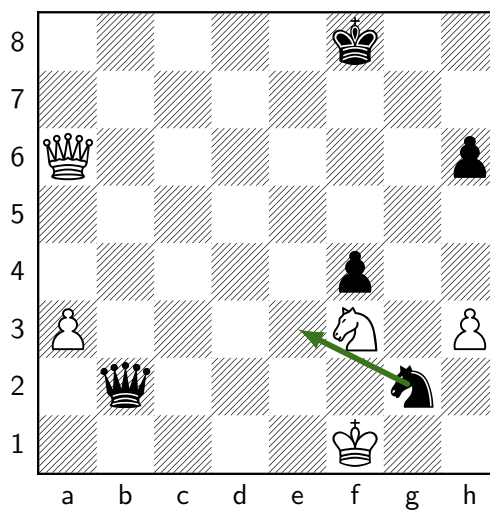


(d)

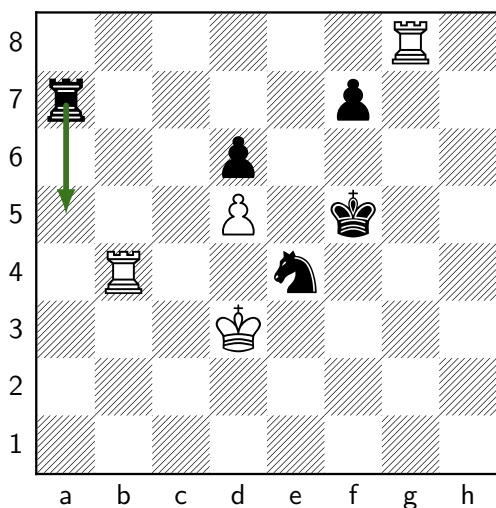
Figure 49: **Move 55, Black: Qxb3+.** (a) Our position; Melanie plays Qxb3+ (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1231` (ply 123) [structural distance $d=56$, QAP obj=35]: the master played Qxg3; under μ our Qa7 plays the role of that game’s Qf6; cross-type roles: our Rf5 as their Pf7, our Nc4 as their pa4; transferred move a2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3865` (ply 122) [structural distance $d=52$, QAP obj=26]: the master played Nxf3; under μ our Qa7 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Rf5 as their Pg4, our Nc4 as their Rb5; transferred move a2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3740` (ply 121) [structural distance $d=56$, QAP obj=24]: the master played Qg4; under μ our Qa7 plays the role of that game’s Qf5; cross-type roles: our nb1 as their bd4, our Rf5 as their pa3; transferred move a2b3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qxb3+ (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



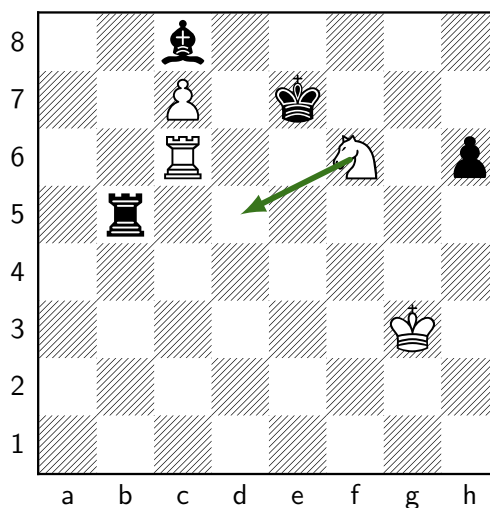
(a)



(b)

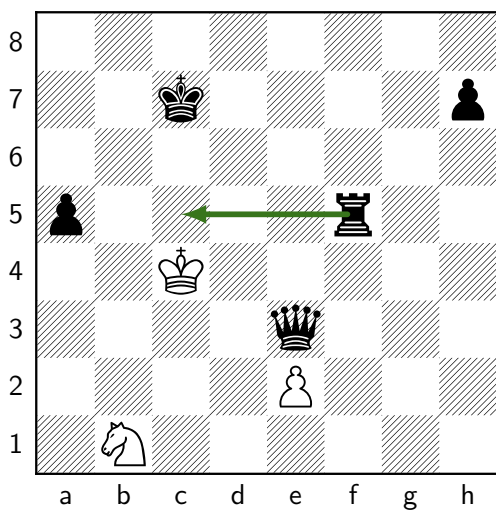


(c)

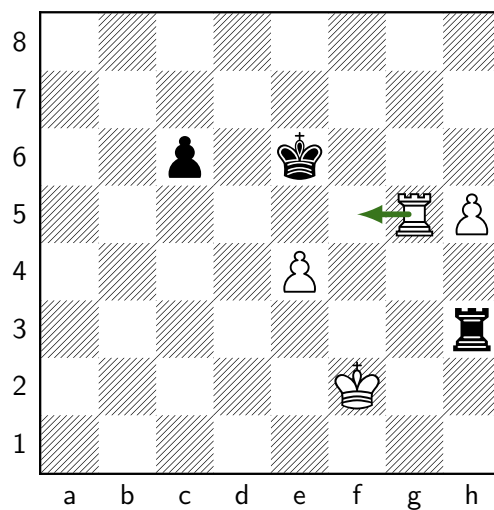


(d)

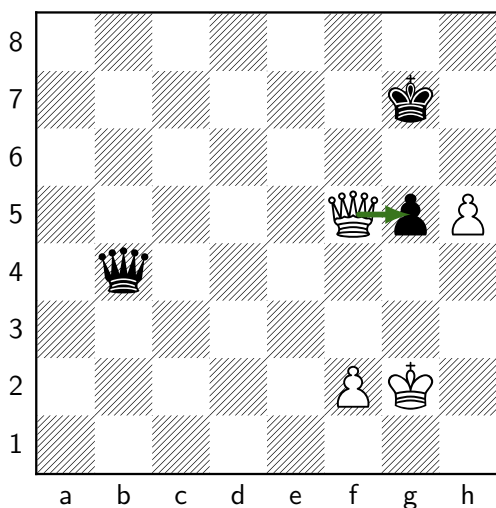
Figure 50: **Move 56, Black: Qe3+.** (a) Our position; Melanie plays Qe3+ (b3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#480` (ply 97) [structural distance $d=29$, QAP obj=28]: the master played Ne3+; under μ our Nc5 plays the role of that game’s Ng7; cross-type roles: our Ph7 as their qa6, our nb1 as their pa3; transferred move c4b2. (c) Retrieved from `ccrl_top_p0000#1856` (ply 109) [structural distance $d=30$, QAP obj=22]: the master played Ra5; under μ our Ph2 plays the role of that game’s Ra2 (a **cross-type** correspondence); cross-type roles: our nb1 as their rg8, our Qb3 as their Pf7; transferred move h7h5. (d) Retrieved from `ccrl_top_p0003#1156` (ply 108) [structural distance $d=29$, QAP obj=21]: the master played Nd5+; under μ our Nc5 plays the role of that game’s Nf6; cross-type roles: our Ph7 as their rb5, our Qb3 as their Rc6; transferred move c4d2. Synthesis: no retrieved analog transferred to Qe3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



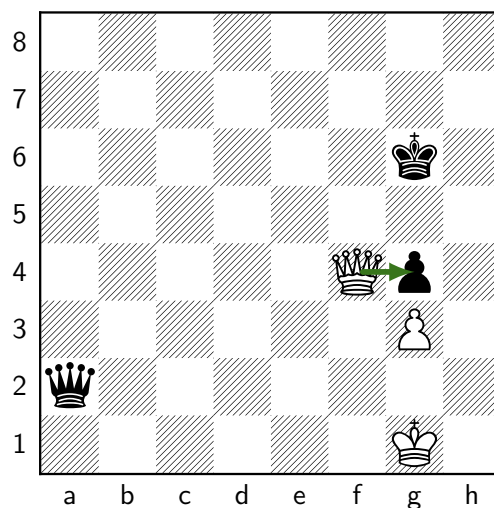
(a)



(b)



(c)



(d)

Figure 51: **Move 57, Black: Rc5#.** (a) Our position; Melanie plays Rc5# (f5c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3961 (ply 106) [structural distance $d=19$, QAP obj=9]: the master played Rf5; under μ our Rf4 plays the role of that game’s Rg5; cross-type roles: our Qe3 as their rh3, our nb1 as their pc6; transferred move f5c5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#290 (ply 98) [structural distance $d=16$, QAP obj=20]: the master played Qxg5+; under μ our Qe6 plays the role of that game’s Qf5; cross-type roles: our nb1 as their qb4; transferred move e3g3. (d) Retrieved from ccr1_top_p0001#2494 (ply 114) [structural distance $d=18$, QAP obj=19]: the master played Qxg4+; under μ our Qe6 plays the role of that game’s Qf4; cross-type roles: our Ph7 as their qa2, our nb1 as their Pg3; transferred move e3h6. Synthesis: 1 retrieved analog transferred to Rc5# (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.